COMPLETE LEARNING SESSION

Emerging Ethics: Technology, AI and Environment - Complete Topic Package

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Emerging Ethics: Technology, AI and Environment

TEN-SESSION CORE PATH • PRACTICE • OPTIONAL ADVANCED • REGISTER NOTES

1. Emerging ethics: visual foundation and essential vocabulary
2. How technology and environmental dilemmas arise and are resolved
3. Indian applications, high-yield facts and close-option traps
4. PYQ entry points, current anchors and Mains theses
5. Digital exclusion, privacy, Puttaswamy and phased DPDP commencement
6. Selectable evidence units for technology and environmental answers
7. Directive decoding and mark-scaled answer architecture
8. Environmental orientations, principles and the Vellore boundary
9. Clausewitz routing: separating environmental ethics from war ethics
10. Study links and complete recent-historical PYQ ownership

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Emerging Ethics: Technology, AI and Environment - Complete Topic Package

Subject: Ethics **Section:** Subject-wide Syllabus **Generation:** learner-v2:g1 **Ownership:** GS-IV Ethics topic; recent official-paper questions retain their exact year, paper and source note. **Source policy:** complete Basic owner first; solved practice before optional Advanced depth; consolidated register notes last.

CURRENT-AFFAIRS ANCHOR

India AI Governance Guidelines and India AI Impact Summit 2026

- FACT: Fact: The India AI Governance Guidelines provide policy guidance through principles, recommendations, an action plan and practical guidance, including voluntary commitments.
- FACT: Fact: The Guidelines cover fairness, accountability, safety, inclusivity, transparency, risk classification, human oversight, testing and grievance redressal.
- FACT: Fact: The India AI Impact Summit was held on 19-20 February 2026 at Bharat Mandapam within a broader 16-20 February programme.
- FACT: Fact: The Summit's official framing promotes people-centric and inclusive AI that protects the planet.
- FACT: Fact: The official Summit page identifies employment disruption, exacerbated bias and accelerating energy consumption as urgent AI challenges.

ANALYSIS: Inference / administrative link: Use the Guidelines to structure purpose, risk classification, testing, human oversight, accountability and grievance safeguards. Use the Summit to connect inclusion and human agency with AI's energy and environmental costs.

ANALYSIS: Limit: Neither source is a comprehensive Indian AI statute, and neither proves that any deployed system is fair, safe or low-carbon. Do not add unsupported impact, participation or emissions metrics.

Official source links:

- https://indiaai.gov.in/article/india-ai-governance-guidelines-empowering-ethical-and-responsible-ai
- https://impact.indiaai.gov.in/about-summit

Source caution: The corrected canonical Topic 13 Basic and Advanced owners control the doctrinal sequence. Official question wording and material case facts come from the stated local GS-IV PDFs or their document-order knowledge exports; routing ledgers establish ownership but do not authorise invented facts. Topic 13 owns only the isolated 2024 Q1(a), 2024 Q2(b), 2025 Q1(a) and 2025 Q2(b) subparts: their paired subparts route respectively to Topic 1, Topic 12, Topic 14 and Topic 12. The full 2024 Q7, 2024 Q12 and 2025 Q8 cases retain Topic 22 method cross-links; Topic 12 is only a corporate cross-link for Q7, and research/drug ethics is the specialist cross-link for Q12. Do not add algorithmic discrimination to 2024 Q7. Do not add degraded forest, city-periphery or court-monitoring facts to 2025 Q8. AI is framed as decision support subject to data, design, deployment and feedback bias controls, explainability, named accountability, meaningful human oversight, contestability, hallucination checks, red-teaming, procurement controls and continuing monitoring. Puttaswamy is the constitutional privacy foundation: use legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards rather than citing a useful aim alone. DPDP commencement is phased under G.S.R. 843(E): specified provisions commenced on 13 November 2025, the Consent Manager tranche begins 13 November 2026, and core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027. Do not call the Act fully operational at the 29 August 2026 cut-off; ethical duties of dignity, autonomy, purpose limitation, minimisation and security apply independently of deferred statutory operation. Rio Principle 15 is the classic environmental precautionary approach. Vellore is an Indian environmental-law authority for sustainable development, precaution and polluter-pays in its tannery-pollution setting; neither Rio nor Vellore is binding AI law. Precaution may guide high-stakes AI only by explicit analogy. Keep precaution ex ante and polluter-pays ex post analytically distinct. Distinguish anthropocentric, biocentric and ecocentric reasons; use environmental justice for burden, voice and remedy; and state CBDR-RC as differentiated responsibility, not zero responsibility. Border-clearance answers require a specific security purpose, EIA and cumulative-risk discipline, community consideration, classified expert review where genuinely necessary, and the least ecologically damaging feasible alternative. The India AI Governance Guidelines and India AI Impact Summit are current official policy anchors, not a comprehensive AI statute or evidence that a particular system is fair, safe or low-carbon.

BASIC LEARNING SESSION

SESSION 1 - Emerging ethics: visual foundation and essential vocabulary

Subject: Ethics | **Tier:** Must-Do (foundation) | **GS Paper:** GS-IV. **Core area:** Emerging ethical challenges from technology (AI, data, automation) and environmental change, as they bear on public administration and business. **Grounded in:** Official UPSC GS-IV syllabus framing (ethics is dynamic, responsive to emerging concerns); *Vellore Citizens' Welfare Forum v. Union of India* (1996) 5 SCC 647; Brundtland Report, *Our Common Future* (1987); Rio Declaration (1992); audited GS-IV PYQs (2024-2025 Mains - AI in administrative decision-making, climate change/human greed, environmental clearance in border areas, the Clausewitz quotation, tech-company GHG case study). FACT: = PYQ/source-grounded | ANALYSIS: = analytical inference | CURRENT: = current anchor. *Companion:* `advanced/13_Emerging-Ethics-Technology-AI-and-Environment.md`.

1. Visual foundation

```
EMERGING DOMAIN         KEY ETHICAL QUESTION                GOVERNANCE RESPONSE NEEDED
AI in administration     Can/should a machine input be a     Human-in-the-loop review,
                          "dependable source" for              explainability, accountability
                          rational decision-making?             for automated recommendations
Data/privacy              Who owns, controls and is            Consent, purpose limitation,
                           accountable for citizen data?          data-protection law compliance
Environmental clearance   How to weigh development against     Precautionary principle,
                           ecological/national-security           proportionality, transparent
                           sensitivity in border areas             environmental impact assessment
Climate change            Is it an ethical failure of          Intergenerational justice,
                           "human greed... in the name of         common-but-differentiated
                           development"? (2024 GS-IV Q2(b))        responsibility
```

Core proposition: ANALYSIS: Emerging ethics extends classical frameworks (duty, consequence, virtue - see `08`) to domains where the *pace of technological/environmental change* outstrips existing rules - requiring anticipatory, precautionary ethical reasoning rather than only retrospective rule application.

2. Essential definitions

Concept	Exam-ready meaning
ANALYSIS: **AI ethics in administration**	The set of concerns (bias, reliability, opacity/explainability, accountability, human oversight and contestability) raised when algorithmic tools inform administrative or judicial decisions.
ANALYSIS: **Algorithmic accountability**	The principle that a human/institution must remain answerable for an AI-assisted decision's outcome - accountability cannot be "delegated" to the algorithm itself.
FACT: **Precautionary principle** (environmental ethics)	Where an activity raises threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation. ANALYSIS: Rio Declaration **Principle 15** (1992), which is its classic statement, actually uses the words "precautionary **approach**".
ANALYSIS: **Intergenerational justice**	The ethical obligation to consider the interests/rights of future generations (who cannot participate in present decision-making) when making resource-use and environmental decisions today.
FACT: **Common but Differentiated Responsibilities (CBDR)**	Rio Declaration Principle 7 and UNFCCC Articles 3.1 and 4.1: all states share responsibility for addressing climate change, but responsibility is differentiated by historical contribution and "respective capabilities". The Paris Agreement (Article 2(2)) expresses it as **CBDR-RC**, "in the light of different national circumstances".

SESSION 2 - How technology and environmental dilemmas arise and are resolved

3. Mechanism: how emerging-ethics dilemmas arise and are resolved

1. **Technology/environmental change outpaces existing rules** - an AI tool or a new industrial process may operate in a genuine regulatory gap, forcing ethical reasoning to substitute for settled law temporarily.
2. **Apply the deontological floor** - does the technology/action violate a non-negotiable right (privacy, non-discrimination, ecological safety) regardless of its efficiency/profitability gain?
3. **Apply consequentialist analysis with a long time-horizon** - climate and AI-bias harms often manifest slowly and diffusely, requiring analysts to resist discounting future/dispersed harm simply because it is less immediately visible than present benefit.
4. **Apply the precautionary principle** where scientific uncertainty is high but potential harm is serious/irreversible (environmental clearance, novel AI deployment in high-stakes decisions).
5. FACT: 2024 GS-IV Q1(a): "application of AI as a dependable source of input for administrative rational decision-making is a debatable issue" - requires "critical examination... from the ethical point of view," directly testing algorithmic accountability and human oversight.
6. FACT: 2024 GS-IV Q2(b): "global warming and climate change are the outcomes of human greed in the name of development" - tests intergenerational justice and the ethics of consumption/growth.
7. FACT: 2025 GS-IV Q2(b): environmental clearance disputes in "ecologically sensitive border areas," examined against "national security" - tests the precautionary principle against strategic/ security imperatives, a genuine multi-value dilemma, not a simple pro-environment answer.

SESSION 3 - Indian applications, high-yield facts and close-option traps

4. Indian applications and examples

- ANALYSIS: An AI-based welfare-eligibility screening tool must retain human review for edge cases and publish its selection criteria to avoid opaque, unaccountable exclusion of genuine beneficiaries.
- FACT: 2025 GS-IV Q2(b)'s border-area environmental-clearance dilemma requires balancing ecological protection, local community rights, and legitimate national-security infrastructure needs - resolvable only through transparent, case-specific environmental impact assessment, not a blanket rule either way.
- ANALYSIS: A data-centre operator's AI-driven emissions growth (see 12) requires verifiable renewable-energy transition commitments, not offsetting through opaque carbon-credit purchases alone.

5. Must-Know Facts for Prelims

- FACT: The precautionary principle is a foundational concept in environmental ethics and law; its classic international statement is Rio Declaration **Principle 15** (1992), which speaks of the "precautionary approach", and it is embedded in India's environmental jurisprudence through *Vellore Citizens' Welfare Forum v. Union of India* (1996).
- FACT: CBDR appears in Rio Declaration Principle 7 and UNFCCC Articles 3.1 and 4.1 (1992); the Paris Agreement's Article 2(2) uses the fuller CBDR-RC formulation.
- ANALYSIS: "Algorithmic accountability" and "explainability" are standard terms in current AI-governance discourse, distinguishing a transparent, auditable AI system from an opaque "black box".

6. UPSC traps

- WRONG: AI-based decision-making is inherently more objective/unbiased than human decision-making. -> AI systems can encode and amplify the biases present in their training data; "objectivity" claims require independent scrutiny, not assumption.
- WRONG: The precautionary principle means all development must stop wherever any risk exists. -> It means proportionate protective measures are justified despite scientific uncertainty for serious/ irreversible risks - not an absolute prohibition on all development.
- WRONG: Climate change is purely a technical/scientific problem with no ethical dimension. -> 2024 GS-IV Q2(b) explicitly frames it as an outcome of "human greed... in the name of development," requiring an ethical, not merely technical, response.

SESSION 4 - PYQ entry points, current anchors and Mains theses

7. PYQ application

- FACT: 2024 GS-IV Q1(a): AI as a "dependable source of input" for administrative decisions - critical examination from the ethical point of view.
- FACT: 2024 GS-IV Q2(b): climate change/global warming as an outcome of human greed - ending it to "protect life and bring equilibrium between society and environment."
- FACT: 2025 GS-IV Q2(b): environmental-clearance disputes in ecologically sensitive border areas, examined against national security.
- FACT: 2024 GS-IV Q7 (Section B case study): tech company's AI-driven data-centre emissions vs profitability/ sustainability (see 12).

8. CURRENT: Current anchors

- CURRENT: **India AI Governance Guidelines (MeitY, released 5 November 2025):** *Enabling Safe and

Trusted AI Innovation* - a policy framework covering fairness, accountability, safety, inclusivity, algorithmic transparency, risk classification, human oversight, testing and grievance redressal. It contains recommendations, an action plan, practical guidance and voluntary commitments; use it for institutional safeguards, **not** as a claim that India has enacted a comprehensive AI statute or that a particular deployed system is fair or safe.

- CURRENT: **India AI Impact Summit, New Delhi:** the headline summit was held on **19-20 February 2026** at Bharat Mandapam, within a broader 16-20 February programme; its official overarching formulation was *Sarvajana Hitaya, Sarvajana Sukhaya*, with "People, Planet, Progress" used as the impact framing that links human-centric AI, inclusion and environmental sustainability.

- CURRENT: **International anchors:** UNESCO's *Recommendation on the Ethics of Artificial Intelligence* (adopted 23 November 2021) and the OECD AI Principles (2019, updated May 2024 - inclusive growth and well-being; human rights and democratic values; transparency and explainability; robustness, security and safety; accountability). Both predate and inform India's own framework.

9. Mains angles

- ANALYSIS: For any AI-ethics question, explicitly name the specific risk (bias, opacity, accountability diffusion, over-reliance/de-skilling of human judgment) rather than giving a generic "AI has pros and cons" answer.

- ANALYSIS: For environment/climate questions, explicitly invoke the precautionary principle and/or CBDR by name, and show the specific multi-value tension (development vs ecology vs security) rather than defaulting to a one-sided "protect environment always" position.

> **Answer thesis:** Emerging technological and environmental challenges require extending classical ethical frameworks with anticipatory tools - algorithmic accountability and the precautionary principle - since existing rules cannot yet fully anticipate harms whose scale, opacity or irreversibility is genuinely novel.

10. Probable questions

- ANALYSIS: **Prelims:** What does the precautionary principle require in conditions of scientific uncertainty?

- ANALYSIS: **Mains (10 marks):** Critically examine the use of AI as a dependable input for administrative decision-making from an ethical standpoint.

- ANALYSIS: **Mains (15 marks):** Examine the ethical dilemmas in granting environmental clearance for development projects in ecologically sensitive border areas, balancing national security and ecological protection.

SESSION 5 - Digital exclusion, privacy, Puttaswamy and phased DPDP commencement

11. Digital gender divide (cross-link)

- ANALYSIS: Internet/technology expansion raises its own gender-equity dilemma: unequal digital access and digital literacy between men and women can widen, not narrow, the substantive-equality gap that formal legal equality alone does not fix - an emerging-ethics dimension of the older gender- inequality question. **Full treatment of the gender-inequality factors and Savitribai Phule's institution-building response (2020 GS-IV Q5(a), historical demand) is in `02_Human-Values-and-Lessons-from-Leaders.md`, Section 8a -** this file's own contribution is flagging the *technology-specific* channel (digital access, algorithmic bias in AI-assisted services) through which the older equity concern re-appears in an emerging-ethics register.

11A. Data/privacy ethics anchor: Puttaswamy and the DPDP Act, 2023

> ANALYSIS: This section is written to be answer-complete for a 10/15-mark question or for the data/privacy dimension of a Section-B case study; full information-governance and DPDP statutory mechanics are owned by `15_Transparency-RTI-and-Information-Sharing.md` and detailed sectoral coverage is in `Science-and-Technology/basic/12_Data-Protection-DPDP-Act-and-Cybersecurity.md` - this file's contribution is the ethical framework (consent, purpose limitation, proportionality) that those files' statutory detail operationalises.

- FACT: **Constitutional anchor:** *Justice K.S. Puttaswamy (Retd.) v. Union of India*, (2017) 10 SCC 1

- a nine-judge Supreme Court bench (24 August 2017) unanimously held that privacy is a fundamental right intrinsic to life and personal liberty under Article 21 and to the freedoms in Part III. ANALYSIS: For an administrative answer, test a State restriction through legality, a legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing, and procedural safeguards. The exact doctrinal formulation developed across the privacy and Aadhaar judgments; do not compress proportionality into a mere statement that the State has a useful aim.

- FACT: **Statutory anchor:** the Digital Personal Data Protection Act, 2023 (DPDP Act) is India's first

dedicated personal-data-protection statute, built around **consent-based processing, purpose limitation** (data used only for the purpose it was collected/consented for), **data minimisation**, and a **Data Protection Board of India** for enforcement.

- CURRENT: **Commencement/status as at 29 August 2026:** G.S.R. 843(E), 13 November 2025, commenced only

specified provisions immediately. The Consent Manager tranche begins on 13 November 2026. Core provisions concerning notice, consent, Data Principal rights, Data Fiduciary and Significant Data Fiduciary duties, transfer restrictions, Board proceedings and penalties begin on 13 May 2027. Thus *Puttaswamy* and the underlying ethical duties of dignity, autonomy, purpose limitation, minimisation and security apply now, while most DPDP operational duties are enacted and notified but not yet in force. Do not describe the Act as fully operational or those future duties as presently enforceable.

- ANALYSIS: **Ethics-law distinction (exam-critical):** Puttaswamy is a *constitutional* holding (privacy as

a fundamental right, subject to a proportionality test on any restriction); the DPDP Act is the *ordinary legislative* mechanism operationalising informational privacy for personal-data processing specifically. An administrator's **ethical** duty to protect a citizen's data (dignity, autonomy, trust) exists independently of, and precedes, the *legal* compliance duty created by a statute's specific, phased provisions - a case study should not treat "the Act is not yet fully in force" as a reason the underlying ethical duty is any weaker.

- ANALYSIS: **Evidence unit:** *Claim:* consent and purpose limitation are the operational ethical core of

data-privacy administration. *Evidence:* the DPDP Act's consent/purpose-limitation/data- minimisation design, read with Puttaswamy's proportionality standard for any state infringement of privacy. *Significance:* gives a two-layer test (is the state aim legitimate and proportionate; is citizen consent/purpose genuinely respected) for any administrative data-use dilemma. *Limitation:* consent can become a formality (long, unread terms) rather than genuine informed consent - an ethical safeguard (plain-language notice, granular consent) is still needed beyond bare statutory compliance.

- ANALYSIS: **Limits:** proportionality review is fact-sensitive and contested case-by-case; the DPDP Act's

phased commencement means some safeguards a candidate might assume are "already enforceable" are not yet operative - always separate the constitutional principle (settled since 2017) from the statute's live commencement status (verify before use).

- WRONG: **UPSC trap:** "The DPDP Act, 2023 has fully replaced Puttaswamy as the source of India's

privacy law." -> Puttaswamy remains the constitutional foundation (privacy as a fundamental right, with a proportionality test); the DPDP Act is a subordinate statutory mechanism for personal-data processing specifically, phased into force, and does not itself amend or supersede the constitutional right.

- FACT: Cross-link: `15_Transparency-RTI-and-Information-Sharing.md` for the DPDP Act's direct amendment of RTI Section 8(1)(j) and the transparency-versus-privacy trade-off in information governance; `22_Case-Study-Method-and-Answer-Architecture.md` Sec. 14.13 for a fully worked eight-element data/privacy case study built on this anchor.

SESSION 6 - Selectable evidence units for technology and environmental answers

12. Selectable named evidence/application units

> Pick 1-2 units per answer; each follows claim -> named evidence/example -> significance -> limitation.

1. *Claim:* Algorithmic accountability requires a human decision-maker to remain answerable.

Evidence: India's AI Governance Guidelines (MeitY, 5 November 2025) "accountability" sutra. *Significance:* gives a current, named institutional anchor for "human-in-the-loop" answers. *Limitation:* guidelines are advisory, not a binding statute - do not claim India has enacted a comprehensive AI law.

2. *Claim:* The precautionary principle justifies proportionate, not absolute, caution. *Evidence:*

Vellore Citizens' Welfare Forum v. Union of India (1996) embedding the principle in Indian environmental jurisprudence. *Significance:* shows the principle already has domestic judicial force, not only an international-declaration status. *Limitation:* proportionate application still requires case-specific scientific assessment - it is not a blanket development veto.

3. *Claim:* CBDR anchors India's climate-negotiation position without denying its own mitigation

responsibility. *Evidence:* UNFCCC Articles 3.1/4.1 and Paris Agreement Article 2(2)'s CBDR-RC formulation. *Significance:* supplies a named, defensible framework for equity-based climate answers. *Limitation:* CBDR is increasingly contested by developed countries seeking universal mitigation commitments - note the framework is a negotiating position, not a settled consensus.

4. *Claim:* Environmental clearance in border areas requires balancing, not a one-sided choice.

Evidence: FACT: 2025 GS-IV Q2(b)'s framing against national security. *Significance:* models a genuine multi-value tension rather than a simple pro-environment stance. *Limitation:* the "balance" still requires a case-specific, transparent EIA process to be more than a rhetorical compromise.

SESSION 7 - Directive decoding and mark-scaled answer architecture

13. Executable directive decoding and answer architecture

Directive word	What is tested	Structural move
Critically examine	AI/technology's benefit and risk both required	Name the specific risk (bias/opacity/accountability diffusion) -> evidence for the benefit claim -> evidence/limitation against -> weighted verdict
Examine the dilemmas	Multi-value tension, not a one-sided position	Name each competing value -> apply the precautionary principle/CBDR where relevant -> balanced, case-specific verdict
Discuss	Balanced treatment with a position	State the emerging concern -> mechanism -> Indian/global example -> limitation -> conclusion

10-mark architecture (~150 words): name the specific emerging-ethics risk/concept precisely -> mechanism -> one named current anchor (Section 8) or example (Section 12) -> one limitation -> one-line conclusion.

Counterpoint and reasoned verdict (template): "[Named technology/environmental mechanism] addresses [X], but [named limitation] means it cannot be relied on alone; the balanced verdict pairs it with [named institutional/legal safeguard]."

SESSION 8 - Environmental orientations, principles and the Vellore boundary

14. Environmental ethics framework: orientations, principles and the Vellore Citizens case

> ANALYSIS: This section closes the environmental-ethics dimension that Sections 1-13 apply but do not name systematically - the three normative *orientations* one can take toward nature, and the operative *principles* (sustainable development, intergenerational equity, precaution, polluter- pays, environmental justice) GS-IV expects an answer to invoke by name.

14A. Three normative orientations toward nature **Semantic table split - Orientation**

- **Orientation:** ANALYSIS: **Anthropocentric**
- **Core claim:** Nature has value chiefly/only insofar as it serves human interests (resources, health, aesthetics, future human welfare).
- **Administrative implication:** Justifies environmental protection *instrumentally* - e.g., protecting a wetland because it recharges groundwater humans need.
- **Main caution:** Can under-protect a species/ecosystem with no direct, provable human-use value.
- **Orientation:** ANALYSIS: **Biocentric**
- **Core claim:** All living organisms have inherent worth, not merely instrumental value to humans; human interests do not automatically override other species' interests.
- **Administrative implication:** Justifies protection of a species *for its own sake* (e.g., an endangered endemic species with no known human-use value).
- **Main caution:** Can be practically difficult to apply when human survival/livelihood needs directly conflict with a non-human organism's interest.
- **Orientation:** ANALYSIS: **Ecocentric**
- **Core claim:** Value attaches to the ecosystem/whole (species, habitats, ecological processes) as an interdependent system, not to individual organisms or humans alone.
- **Administrative implication:** Justifies protecting ecological processes and habitats as a unit (e.g., a river's entire flow regime, not just the fish in it).
- **Main caution:** Can be invoked to block *any* human use of an ecosystem, including sustainable/traditional community use, if applied without proportionality.
- ANALYSIS: **Stewardship caution:** a "stewardship" framing (humans as trustees, not owners, of nature -

echoing Gandhian trusteeship, see `06`) is a workable *administrative* middle path between pure anthropocentrism and pure ecocentrism, but it still centres human responsibility/agency - do not present it as identical to biocentrism/ecocentrism in an answer that specifically asks the three orientations to be distinguished.

- ANALYSIS: **Deep-ecology caution:** "deep ecology" (Arne Naess's term for a biocentric/ecocentric position

that rejects anthropocentrism as the root cause of the environmental crisis) is a genuine named philosophical position, but a GS-IV answer should present it as *one* school of environmental thought, not as the uncontested correct one - India's own constitutional and statutory framework (Article 48A/51A(g), the Environment (Protection) Act, 1986) is textually anthropocentric-cum- stewardship in structure (protecting the environment "for the sake of" present and future generations), not deep-ecological in the strict Naess sense; over-claiming a single orientation as "correct" is itself a trap (Sec. 14C).

14B. Operative principles

Concept	Exam-ready meaning
FACT: **Sustainable development**	Development that "meets the needs of the present without compromising the ability of future generations to meet their own needs" (Brundtland Report, *Our Common Future*, 1987) - the standard bridging concept between development and ecological limits.
ANALYSIS: **Intergenerational equity**	The ethical obligation to leave future generations no worse off in their capacity to meet their needs than the present generation - the temporal extension of "public interest" beyond those currently living (see `08` on long-time-horizon consequentialism).
FACT: **Precautionary approach/principle**	Where an activity threatens serious or irreversible damage, lack of full scientific certainty is not a reason to postpone cost-effective preventive measures (Rio Declaration Principle 15, 1992 - see Section 2 above for the "approach" vs "principle" wording caution).
FACT: **Polluter-pays principle**	The party causing pollution bears the cost of remedying/compensating the resulting environmental harm, rather than the cost falling on the state/public exchequer or affected citizens by default - applied by the Supreme Court in *Vellore Citizens' Welfare Forum* (below) as part of Indian environmental law.
ANALYSIS: **Environmental justice**	The claim that environmental burdens (pollution exposure, displacement, resource loss) and benefits (clean environment, compensation, resettlement) must be distributed fairly, with particular attention to disproportionate burdens on the poor, tribal and marginalised communities who are least able to bear or contest them.

14C. *Vellore Citizens' Welfare Forum v. Union of India* - significance and limits

- FACT: **Citation and holding:** (1996) 5 SCC 647, decided 28 August 1996. Leather tanneries in Tamil

Nadu's Vellore district discharged untreated effluent into the Palar river, contaminating agricultural land and drinking water. The Supreme Court held that **sustainable development**, the **precautionary principle** and the **polluter-pays principle** are part of the environmental law of India and part of Indian environmental jurisprudence. It directed closure of non-complying tanneries pending effluent-treatment compliance and required an authority mechanism under the Environment (Protection) Act to assess compensation for ecological damage and affected persons.

- ANALYSIS: **What it correctly stands for:** it is the leading Indian judicial anchor for treating

sustainable development, precaution and polluter-pays as *justiciable* legal principles (not merely policy aspirations) enforceable via Article 21 and the Supreme Court's writ jurisdiction - a genuinely strong, citable precedent for "environmental principles have legal force in India" answers.

- ANALYSIS: **Its limits - do not over-claim:** (a) it is a **specific-industry, specific-river** judgment

(tannery effluent on the Palar) resolved through case-specific directions and a compensation- authority mechanism, not a general environmental code covering every future dispute; (b) its practical setting now includes the specialised **National Green Tribunal** (constituted 2010), which applies sustainable development, precaution and polluter-pays under NGT Act Section 20; the NGT does not displace the Supreme Court's constitutional jurisdiction or erase *Vellore*; (c) the judgment's balancing of industry/livelihood against ecological harm is itself an anthropocentric-leaning application (protecting human health/agriculture from pollution) rather than an ecocentric one - a nuance worth naming if a question specifically tests the orientations of Section 14A; (d) it does not, by itself, resolve every precautionary-principle application (e.g., novel AI deployment, Section 1-9 above) - it is the domestic *environmental* anchor specifically, not a general-purpose precautionary-principle authority for all emerging- ethics domains.

14D. UPSC traps (environmental ethics)

- WRONG: Ecocentrism is the only defensible environmental-ethics position and anthropocentrism is simply wrong. -> GS-IV rewards naming the tension between orientations and applying the one the scenario calls for, not declaring one orientation universally correct; India's own legal framework is predominantly a sustainable-development/stewardship synthesis, not strict deep ecology.

- WRONG: Sustainable development means environment always yields to development, or always overrides it. -> Sustainable development is precisely the attempt to hold *both* present development needs and future-generation ecological capacity together - treating it as a one-sided override in either direction misstates the Brundtland definition.

- WRONG: *Vellore Citizens' Welfare Forum* is a general environmental statute/code. -> It is a Supreme Court judgment resolving a specific tannery-pollution dispute that established precaution and polluter-pays as part of Indian law; ongoing enforcement now runs substantially through the NGT.

- WRONG: The precautionary principle and the polluter-pays principle are the same thing. -> Precaution governs *ex ante* decision-making under scientific uncertainty (should the activity proceed at all, and with what safeguards); polluter-pays governs *ex post* liability for harm actually caused - a strong answer keeps the two analytically separate even when citing the same case for both.

14E. 10/20-mark architecture integration (environmental ethics) **10-mark architecture (~150 words):** name the specific principle/orientation the question tests (precaution, polluter-pays, intergenerational equity, or an anthropocentric/biocentric/ecocentric framing) -> define it precisely -> cite *Vellore Citizens' Welfare Forum* (1996) as the domestic judicial anchor where relevant, stating its actual holding, not a generalised claim -> one Indian example/limitation -> one-line conclusion naming the safeguard (NGT enforcement, EIA process, or a named institutional review).

20-mark/case-study architecture (~250 words): facts/stakeholders (including future-generation and non-human/ecological stakeholders explicitly, per Sec. 14A) -> constraints (scientific uncertainty, development pressure, livelihood dependence) -> options weighted through the precautionary and polluter-pays principles, not a blanket "protect environment" or "allow development" position -> decision naming which orientation/principle was decisive and why -> the 22-style implementation/communication and residual-risk elements, with the residual ecological/ compliance risk named explicitly rather than assumed resolved.

SESSION 9 - Clausewitz routing: separating environmental ethics from war ethics

15. Clausewitz and war ethics - non-owner routing note (2025 GS-IV Q2(a))

> CURRENT: The audited 2025 GS-IV Q2(a) demand - "Clausewitz - war as diplomacy by other means" - appears beside Q2(b) in the printed paper, but it is **not owned by Topic 13**. The full Just War/IHL/proportionality framework needed to answer it is owned by `12_Corporate-Governance-and-International-Ethics.md`, Section 11 ("War ethics and international aid ethics"). Topic 13 owns only Q2(b), the environmental-clearance dilemma.

- ANALYSIS: **Attribution/context caution:** the phrase commonly rendered "war is the continuation of policy [or "politics"] by other means" is a paraphrase of Carl von Clausewitz's *On War* (*Vom Kriege*, published posthumously in 1832 by his widow, Marie von Clausewitz); Clausewitz wrote in German, and English renderings vary between "policy" and "politics" depending on the translator (the standard modern Howard-Paret translation uses "policy") - an answer should treat this as a paraphrased translation of a 19th-century Prussian military-theory text, not a verbatim English quotation with one fixed canonical wording, and should not present it as an ethics treatise in the first instance - it is strategic theory that GS-IV repurposes for an ethics question about the *justification* of war as an instrument of state policy.

- ANALYSIS: **What an ethics answer must add beyond the attribution:** Clausewitz's claim is a *descriptive/strategic* one (war is an extension of political intercourse, not a UN-Charter-era ethical claim about when war is justified) - a GS-IV answer should pair it with the *normative* Just War framework (sovereignty/*jus ad bellum*, civilian protection, proportionality, humanitarian neutrality) fully set out in `12`'s Section 11, rather than treating the quotation alone as a complete ethical position. This file's contribution is the attribution caution and this explicit cross-link; `12` remains the substantive owner for the ethical-framework content the question requires.

SESSION 10 - Study links and complete recent-historical PYQ ownership

16. Study links

- FACT: Advanced companion: `advanced/13_Emerging-Ethics-Technology-AI-and-Environment.md`.
- FACT: `12_Corporate-Governance-and-International-Ethics.md` - the corporate AI/environment case study,

and Section 11's Just War/international-aid-ethics framework for the Clausewitz routing (Sec. 15).

- FACT: `08_Moral-Theories-Deontology-Consequentialism-Virtue-Ethics.md` - theory application to novel dilemmas.
- FACT: `06_Indian-Moral-Thinkers-and-Philosophers.md` - Gandhian trusteeship as the stewardship anchor (Sec. 14A).
- FACT: `15_Transparency-RTI-and-Information-Sharing.md` - the DPDP Act's amendment of RTI Section

8(1)(j) and information-governance detail for the data/privacy anchor (Sec. 11A).

- FACT: `22_Case-Study-Method-and-Answer-Architecture.md` Sec. 14.13 - the fully worked data/privacy

case study built on Sec. 11A's Puttaswamy/DPDP anchor.

- FACT: `Science-and-Technology/basic/09_Artificial-Intelligence-Governance-and-IndiaAI.md` - AI

governance/policy detail (avoid duplication here).

- FACT: `Environment-and-Ecology/basic/16_Environmental-Impact-Assessment-and-NGT.md` - EIA process/law

detail and the National Green Tribunal's current enforcement role (Sec. 14C).

- FACT: `Environment-and-Ecology/basic/19_UNFCCC-COP-Kyoto-Paris-Agreement.md` - CBDR and climate-law detail (avoid duplication here).

Recent PYQ Integration (2024-2025)

> **Status:** 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS3-GS4-2024-2025.md`.

- **Years represented:** 2024, 2025
- **Paper(s):** GS-IV
- **Routed question demands:** 7 isolated subparts/cases

Semantic table split - Year

- **Year:** 2024
- **Paper:** GS-IV
- **Q:** 1(a)
- **PYQ demand (neutral rendering):** AI as a dependable input for administrative rational decision-making
- **Directive / format:** Section A theory · 10 marks · 150 words
- **Source status:** Topic 13 primary
- **Owner requirement:** Prepare benefits, bias/reliability risks, human oversight, reasons, appeal and a qualified verdict.
- **Year:** 2024
- **Paper:** GS-IV
- **Q:** 2(b)
- **PYQ demand (neutral rendering):** Global warming and climate change as outcomes of greed in the name of development

- **Directive / format:** Section A theory · 10 marks · 150 words
- **Source status:** Topic 13 primary
- **Owner requirement:** Apply virtue ethics, intergenerational justice, sustainable development and practical restraint.
- **Year:** 2024
- **Paper:** GS-IV
- **Q:** 7
- **PYQ demand (neutral rendering):** ABC Incorporated: AI/data-centre energy growth, GHG increase, net-zero pledge and competitive pressure
- **Directive / format:** Section B case study · 20 marks · 250 words
- **Source status:** Topic 22 method; Topic 13 substance; Topic 12 cross-link
- **Owner requirement:** Preserve the official emissions and competition facts; do not invent an algorithmic-discrimination strand.
- **Year:** 2024
- **Paper:** GS-IV
- **Q:** 12
- **PYQ demand (neutral rendering):** Dr. Srinivasan: adverse-result suppression, informed consent and patented-compound issues in viral-drug trials
- **Directive / format:** Section B case study · 20 marks · 250 words
- **Source status:** Topic 22 method; Topic 13 data-ethics cross-link
- **Owner requirement:** Preserve every research-ethics fact and use a biotechnology/research-ethics cross-link.
- **Year:** 2025
- **Paper:** GS-IV
- **Q:** 1(a)
- **PYQ demand (neutral rendering):** Ethical dilemmas created by social media in the digital age
- **Directive / format:** Section A theory · 10 marks · 150 words
- **Source status:** Topic 13 primary
- **Owner requirement:** Isolate social-media ethics; Q1(b) constitutional morality belongs to Topic 14.
- **Year:** 2025
- **Paper:** GS-IV
- **Q:** 2(b)
- **PYQ demand (neutral rendering):** Environmental clearance in ecologically sensitive border areas, keeping national security in mind
- **Directive / format:** Section A theory · 10 marks · 150 words
- **Source status:** Topic 13 primary
- **Owner requirement:** Isolate environmental clearance; Q2(a) Clausewitz belongs to Topic 12.
- **Year:** 2025
- **Paper:** GS-IV
- **Q:** 8
- **PYQ demand (neutral rendering):** Housing for homeless/EWS groups proposed on ecologically sensitive forest land
- **Directive / format:** Section B case study · 20 marks · 250 words
- **Source status:** Topic 22 method; Topic 13 substance
- **Owner requirement:** Preserve old trees, medicinal plants, biodiversity, ecosystem services and tribal/nomadic livelihoods; do not add degradation or court-monitoring facts.

What this owner must now support

- 2024 Q1(a): AI as a dependable input for administrative rational decision-making
- 2024 Q2(b): global warming, climate change and human greed
- 2024 Q7: ABC Incorporated's AI/data-centre emissions and net-zero dilemma
- 2024 Q12: research-data integrity and informed-consent cross-link in Dr. Srinivasan's case
- 2025 Q1(a): ethical dilemmas created by social media
- 2025 Q2(b): environmental clearance in ecologically sensitive border areas
- 2025 Q8: forest ecology, housing and human-dignity case

> This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md`.

- **Years represented:** 2018, 2020, 2021, 2022, 2023
- **Paper(s):** GS-IV
- **Routed question demands:** 7

Semantic table split - Year

- **Year:** 2018
- **Paper:** GS-IV
- **Q:** 5
- **PYQ demand (neutral rendering):** (a) government dam construction in forest valley inhabited by ethnic communities - rational policy for unforeseen contingencies; (b) process of resolving ethical dilemmas in public administration
- **Directive / format:** Suggest/Explain · 10 + 10 marks · 150 words each
- **Source status:** Routed to owning Ethics topic
- **Owner requirement:** Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
- **Year:** 2018
- **Paper:** GS-IV
- **Q:** 10
- **PYQ demand (neutral rendering):** Corporate chemical factory ordered closed after pollution protests and public agitation; closure causes unemployment in workers and ancillary units; address as senior officer
- **Directive / format:** Case study · 20 marks · 250 words
- **Source status:** Case routed to Ethics case-study method
- **Owner requirement:** Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
- **Year:** 2020
- **Paper:** GS-IV
- **Q:** 5
- **PYQ demand (neutral rendering):** (a) main factors for gender inequality in India and Savitribai Phule contribution; (b) internet expansion instilling cultural values in conflict with traditional values
- **Directive / format:** Discuss · 10 + 10 marks · 150 words each
- **Source status:** Routed to owning Ethics topic

- **Owner requirement:** Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
- **Year:** 2021
- **Paper:** GS-IV
- **Q:** 2
- **PYQ demand (neutral rendering):** (a) digital technology as reliable input for rational decision making - critically evaluate; (b) innovativeness and creativity in resolving ethical dilemmas
- **Directive / format:** Critically evaluate / Discuss · 10 + 10 marks · 150 words each
- **Source status:** Routed to owning Ethics topic
- **Owner requirement:** Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
- **Year:** 2022
- **Paper:** GS-IV
- **Q:** 4
- **PYQ demand (neutral rendering):** (a) good governance and e-governance initiatives for beneficiaries; (b) ethical issues in online methodology affecting vulnerable sections of society
- **Directive / format:** Discuss · 10 + 10 marks · 150 words each
- **Source status:** Routed to owning Ethics topic
- **Owner requirement:** Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
- **Year:** 2022
- **Paper:** GS-IV
- **Q:** 12
- **PYQ demand (neutral rendering):** Case on Environment Pollution Control Board officer enforcing industrial environmental compliance despite resistance and threats
- **Directive / format:** Case study · 20 marks · 250 words
- **Source status:** Case routed to Ethics case-study method
- **Owner requirement:** Apply stakeholders, dilemmas, options, justification, implementation and safeguards.
- **Year:** 2023
- **Paper:** GS-IV
- **Q:** 12
- **PYQ demand (neutral rendering):** Case on senior government official managing son's cyberbullying and posting a counter-video on social media
- **Directive / format:** Case study · 20 marks · 250 words
- **Source status:** Case routed to Ethics case-study method
- **Owner requirement:** Apply stakeholders, dilemmas, options, justification, implementation and safeguards.

What this owner must now support

- (a) government dam construction in forest valley inhabited by ethnic communities - rational policy for unforeseen contingencies; (b) process of resolving ethical dilemmas in public administration
- Corporate chemical factory ordered closed after pollution protests and public agitation; closure causes unemployment in workers and ancillary units; address as senior officer
- (a) main factors for gender inequality in India and Savitribai Phule contribution; (b) internet expansion instilling cultural values in conflict with traditional values
- (a) digital technology as reliable input for rational decision making - critically evaluate; (b) innovativeness and creativity in resolving ethical dilemmas

- (a) good governance and e-governance initiatives for beneficiaries; (b) ethical issues in online methodology affecting vulnerable sections of society
- Case on Environment Pollution Control Board officer enforcing industrial environmental compliance despite resistance and threats
- Case on senior government official managing son's cyberbullying and posting a counter-video on social media

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

BASIC MCQS / REMEDIATION

MCQ 1

A welfare department treats a model score as conclusive and refuses to identify any responsible officer. Which principle has been displaced? Which source-grounded ethical principle most precisely explains the case?

A. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.

B. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.

C. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.

D. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.

Answer: A

Explanation: AI is decision support, not an accountability substitute is the controlling principle. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 2

A licensing officer checks an AI risk flag against the record and gives independent reasons. Which proper role for AI is illustrated? Which source-grounded ethical principle most precisely explains the case?

A. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.

B. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.

C. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.

D. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.

Answer: B

Explanation: AI is decision support, not an accountability substitute is the controlling principle. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 3

A fraud model trained only on urban cases repeatedly misclassifies rural claims. At which bias stage should the diagnosis begin? Which source-grounded ethical principle most precisely explains the case?

A. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.

B. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.

C. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.

D. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.

Answer: C

Explanation: Bias can enter data, design, deployment and feedback is the controlling principle. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 4

A policing model sends more patrols where it earlier predicted risk, then treats the resulting arrests as proof. Which loop is operating? Which source-grounded ethical principle most precisely explains the case?

A. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.

B. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.

C. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.

D. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.

Answer: D

Explanation: Bias can enter data, design, deployment and feedback is the controlling principle. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 5

A pension applicant receives only a model version number after rejection. Which ethical safeguard remains unsatisfied? Which source-grounded ethical principle most precisely explains the case?

A. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.

B. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.

C. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.

D. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.

Answer: A

Explanation: Explainability must be useful to the affected person is the controlling principle. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 6

An agency states the decisive data, rule, uncertainty and appeal route in plain language. Which concept is being operationalised? Which source-grounded ethical principle most precisely explains the case?

A. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.

B. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.

C. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.

D. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.

Answer: B

Explanation: Explainability must be useful to the affected person is the controlling principle. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 7

A junior clerk must approve every model output within ten seconds and cannot alter it. Why is the human-in-the-loop claim hollow? Which source-grounded ethical principle most precisely explains the case?

A. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.

B. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.

C. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.

D. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.

Answer: C

Explanation: Human oversight must be meaningful is the controlling principle. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 8

A medical board can pause, investigate and reverse a high-risk model recommendation. Which form of oversight is present? Which source-grounded ethical principle most precisely explains the case?

A. AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.

B. Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.

C. Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.

D. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.

Answer: D

Explanation: Human oversight must be meaningful is the controlling principle. Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 9

A department blames a vendor after an automated exclusion although no official owned validation. Which governance failure is central? Which source-grounded ethical principle most precisely explains the case?

A. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.

B. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.

C. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.

D. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.

Answer: A

Explanation: Accountability requires an auditable responsibility chain is the controlling principle. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 10

A procurement contract preserves logs, assigns incident duties and permits independent audit. Which accountability architecture is strengthened? Which source-grounded ethical principle most precisely explains the case?

A. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.

B. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.

C. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.

D. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.

Answer: B

Explanation: Accountability requires an auditable responsibility chain is the controlling principle. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 11

A scholarship applicant can submit corrected records but no one may reconsider the model result. Which element remains absent? Which source-grounded ethical principle most precisely explains the case?

A. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.

B. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.

C. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.

D. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.

Answer: C

Explanation: Contestability gives affected persons an effective remedy is the controlling principle. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 12

A citizen receives notice, a simple appeal channel and reasoned human review. Which safeguard is demonstrated? Which source-grounded ethical principle most precisely explains the case?

A. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.

B. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.

C. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.

D. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.

Answer: D

Explanation: Contestability gives affected persons an effective remedy is the controlling principle. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 13

Experienced inspectors stop examining unusual cases because the dashboard is usually right. Which paired risks arise? Which source-grounded ethical principle most precisely explains the case?

A. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.

B. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.

C. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.

D. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.

Answer: A

Explanation: Automation bias and de-skilling are institutional risks is the controlling principle. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 14

A department requires periodic manual sampling and records justified departures from model advice. Which risks is it trying to control? Which source-grounded ethical principle most precisely explains the case?

A. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.

B. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.

C. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.

D. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.

Answer: B

Explanation: Automation bias and de-skilling are institutional risks is the controlling principle. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 15

An officer files a disciplinary order using model-generated cases without checking them. Which reliability failure is decisive? Which source-grounded ethical principle most precisely explains the case?

A. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.

B. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.

C. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.

D. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.

Answer: C

Explanation: Hallucination and reliability require verification is the controlling principle. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 16

A legal cell verifies every generated citation against official records before use. Which control addresses hallucination? Which source-grounded ethical principle most precisely explains the case?

A. Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.

B. Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.

C. Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.

D. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.

Answer: D

Explanation: Hallucination and reliability require verification is the controlling principle. Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 17

A vendor passes one demonstration and then changes its model without notice. Which procurement controls were missing? Which source-grounded ethical principle most precisely explains the case?

A. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.

B. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.

C. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.

D. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

Answer: A

Explanation: Procurement must include red-teaming and monitoring is the controlling principle. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 18

A department tests adversarial cases before launch and monitors drift after deployment. Which lifecycle approach is illustrated? Which source-grounded ethical principle most precisely explains the case?

A. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.

B. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.

C. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

D. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.

Answer: B

Explanation: Procurement must include red-teaming and monitoring is the controlling principle. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 19

A forged video of a district officer triggers panic before verification. Which ethical value is most directly attacked? Which source-grounded ethical principle most precisely explains the case?

A. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.

B. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.

C. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.

D. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

Answer: C

Explanation: Deepfakes attack authenticity and public trust is the controlling principle. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 20

Authorities preserve the original, verify provenance and issue a prompt correction without suppressing criticism. Which balanced response is shown? Which source-grounded ethical principle most precisely explains the case?

A. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.

B. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

C. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.

D. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.

Answer: D

Explanation: Deepfakes attack authenticity and public trust is the controlling principle. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 21

A platform rewards inflammatory falsehood because engagement rises. Which design-linked ethical problem is present? Which source-grounded ethical principle most precisely explains the case?

A. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.

B. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.

C. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.

D. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

Answer: A

Explanation: Social media combines misinformation, manipulation and privacy harms is the controlling principle. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 22

A public agency counters a rumour with verified facts while protecting children's identities. Which competing values are being balanced? Which source-grounded ethical principle most precisely explains the case?

A. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.

B. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.

C. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

D. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.

Answer: B

Explanation: Social media combines misinformation, manipulation and privacy harms is the controlling principle. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 23

A village has mobile coverage but elderly pensioners cannot navigate an English-only portal. Which divide persists? Which source-grounded ethical principle most precisely explains the case?

A. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.

B. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.

C. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

D. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.

Answer: C

Explanation: The digital divide is about capability, not connectivity alone is the controlling principle. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 24

A service adds assisted access, local-language design and an offline channel. Which ethical principle is advanced? Which source-grounded ethical principle most precisely explains the case?

A. Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.

B. Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.

C. Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.

D. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

Answer: D

Explanation: The digital divide is about capability, not connectivity alone is the controlling principle. Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 25

A health database is reused for unrelated profiling because the facts were once disclosed voluntarily. Which understanding of privacy rejects this reasoning? Which source-grounded ethical principle most precisely explains the case?

A. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.

B. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.

C. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.

D. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.

Answer: A

Explanation: Privacy protects dignity, autonomy and contextual control is the controlling principle. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 26

An agency limits reuse to the stated service purpose and permits correction. Which underlying values are protected? Which source-grounded ethical principle most precisely explains the case?

A. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.

B. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.

C. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.

D. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.

Answer: B

Explanation: Privacy protects dignity, autonomy and contextual control is the controlling principle. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 27

A department cites convenience but identifies no law or narrower alternative before mass tracking. Which constitutional test is incomplete? Which source-grounded ethical principle most precisely explains the case?

A. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.

B. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.

C. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.

D. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.

Answer: C

Explanation: Puttaswamy requires the full proportionality inquiry is the controlling principle. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 28

A scheme has legal authority, a legitimate aim, narrow collection, safeguards and review. Which analytical framework supports assessment? Which source-grounded ethical principle most precisely explains the case?

A. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.

B. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.

C. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.

D. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.

Answer: D

Explanation: Puttaswamy requires the full proportionality inquiry is the controlling principle. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 29

An answer says every DPDP duty was enforceable from November 2025. Which chronology corrects it? Which source-grounded ethical principle most precisely explains the case?

A. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.

B. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.

C. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.

D. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.

Answer: A

Explanation: DPDP commencement is phased, not fully operational is the controlling principle. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 30

An administrator protects data ethically now while distinguishing future statutory duties. Which law-ethics distinction is shown? Which source-grounded ethical principle most precisely explains the case?

A. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.

B. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.

C. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.

D. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.

Answer: B

Explanation: DPDP commencement is phased, not fully operational is the controlling principle. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 31

A benefits app collects contacts and location unrelated to eligibility under one bundled checkbox. Which safeguards fail? Which source-grounded ethical principle most precisely explains the case?

A. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.

B. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.

C. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.

D. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.

Answer: C

Explanation: Consent must be purpose-bound and security-backed is the controlling principle. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 32

A service requests only necessary fields, explains use and deletes them when no longer needed. Which data ethic is applied? Which source-grounded ethical principle most precisely explains the case?

A. A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.

B. G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.

C. Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.

D. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.

Answer: D

Explanation: Consent must be purpose-bound and security-backed is the controlling principle. Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 33

A company reports renewable electricity but ignores cooling water and discarded accelerators. Why is its footprint account incomplete? Which source-grounded ethical principle most precisely explains the case?

A. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.

B. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.

C. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.

D. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.

Answer: A

Explanation: AI has energy, water, hardware and e-waste costs is the controlling principle. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 34

A data centre measures operational energy, water, embodied hardware and e-waste. Which lifecycle view is demonstrated? Which source-grounded ethical principle most precisely explains the case?

A. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.

B. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.

C. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.

D. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.

Answer: B

Explanation: AI has energy, water, hardware and e-waste costs is the controlling principle. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 35

A housing plan ignores ecosystem services because present welfare always prevails. Which principle corrects that one-sided claim? Which source-grounded ethical principle most precisely explains the case?

A. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.

B. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.

C. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.

D. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.

Answer: C

Explanation: Sustainable development is a two-generation bridge is the controlling principle. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 36

A project redesign protects livelihoods while retaining ecological capacity. Which bridging concept supports it? Which source-grounded ethical principle most precisely explains the case?

A. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.

B. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.

C. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.

D. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.

Answer: D

Explanation: Sustainable development is a two-generation bridge is the controlling principle. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 37

A government discounts irreversible groundwater loss because future residents cannot object. Which ethical duty is neglected? Which source-grounded ethical principle most precisely explains the case?

A. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.

B. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.

C. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.

D. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.

Answer: A

Explanation: Intergenerational equity represents absent future citizens is the controlling principle. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 38

A policy applies a long time horizon and preserves ecological options. Which justice principle is operating? Which source-grounded ethical principle most precisely explains the case?

A. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.

B. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.

C. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.

D. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.

Answer: B

Explanation: Intergenerational equity represents absent future citizens is the controlling principle. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 39

A note calls Vellore a directly enforceable AI statute. Which doctrinal limit must be restored? Which source-grounded ethical principle most precisely explains the case?

A. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.

B. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.

C. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.

D. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.

Answer: C

Explanation: Precaution may guide high-stakes AI only by analogy is the controlling principle. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 40

A department pilots and stress-tests an uncertain high-impact AI system before scale-up. Which analogous reasoning supports caution? Which source-grounded ethical principle most precisely explains the case?

A. Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.

B. Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.

C. AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.

D. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.

Answer: D

Explanation: Precaution may guide high-stakes AI only by analogy is the controlling principle. Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 41

A factory pays a fee and claims unlimited discharge is now ethical. Which misunderstanding is present? Which source-grounded ethical principle most precisely explains the case?

A. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.

B. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.

C. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.

D. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

Answer: A

Explanation: Polluter-pays allocates prevention and remediation costs is the controlling principle. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 42

A regulator requires cleanup, victim compensation and preventive upgrades from the polluter. Which principle is applied? Which source-grounded ethical principle most precisely explains the case?

A. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.

B. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.

C. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

D. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.

Answer: B

Explanation: Polluter-pays allocates prevention and remediation costs is the controlling principle. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 43

A clean-energy project displaces a tribal community without consultation while distant users gain. Which justice lens reveals the burden? Which source-grounded ethical principle most precisely explains the case?

A. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.

B. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.

C. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.

D. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

Answer: C

Explanation: Environmental justice tests distribution and voice is the controlling principle. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 44

A clearance process includes affected communities, fair compensation and accessible remedy. Which ethical concern is addressed? Which source-grounded ethical principle most precisely explains the case?

A. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.

B. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

C. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.

D. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.

Answer: D

Explanation: Environmental justice tests distribution and voice is the controlling principle. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 45

A wetland is protected solely for drinking-water recharge. Which orientation supplies the stated reason? Which source-grounded ethical principle most precisely explains the case?

A. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.

B. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.

C. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.

D. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

Answer: A

Explanation: Anthropocentric, biocentric and ecocentric reasons differ is the controlling principle. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 46

A river's flow regime is protected as an interdependent ecological whole. Which orientation is most precise? Which source-grounded ethical principle most precisely explains the case?

A. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.

B. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.

C. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

D. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.

Answer: B

Explanation: Anthropocentric, biocentric and ecocentric reasons differ is the controlling principle. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 47

A climate answer says developing states have no duties. Which differentiated principle rejects that conclusion? Which source-grounded ethical principle most precisely explains the case?

A. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.

B. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.

C. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

D. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.

Answer: C

Explanation: CBDR-RC and EIA require differentiated, least-damaging choices is the controlling principle. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

MCQ 48

A border road uses a narrower alignment after rigorous assessment while protecting genuine security information. Which balanced method is shown? Which source-grounded ethical principle most precisely explains the case?

A. Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.

B. Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.

C. Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.

D. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

Answer: D

Explanation: CBDR-RC and EIA require differentiated, least-damaging choices is the controlling principle. CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative. The distractors are related propositions, but they do not identify the decisive mechanism in this case.

PYQS AND ANSWER PRACTICE

Solved PYQ 1 - 2018 - 20 marks

Question: GS-IV Q10: A big corporate house is engaged in manufacturing industrial chemicals on a large scale. It proposes to set up an additional unit. Many States rejected its proposal due to detrimental effect on the environment. But one State government acceded to the request and permitted the unit close to a city, brushing aside all opposition. The unit was set up 10 years ago and was in full swing till recently. The pollution caused by the industrial effluents was affecting the land, water and crops in the area. It was also causing serious health problems to human beings and animals. This gave rise to a series of agitations demanding the closure of the plant. In a recent agitation thousands of people took part, creating a law and order problem necessitating stern police action. Following the public outcry, the State government ordered the closure of the factory. The closure of the factory resulted in the unemployment of not only those workers who were engaged in the factory but also those who were working in the ancillary units. It also very badly affected those industries which depended on the chemicals manufactured by it. As a senior officer entrusted with the responsibility of handling this issue, how are you going to address it? (Answer in 250 words)

Source / ownership: Exact full case verified against books\more_previous_papers\GENERAL-STUDIES-PAPER-IV.pdf, page 9. Topic 13 owns pollution, precaution, polluter-pays and environmental-justice substance; Topic 22 supplies the general case-study method.

Model solution

The officer must reject the false choice between uncontrolled production and abrupt livelihood destruction. Stakeholders include polluted communities, workers and their families, ancillary firms, consumers, the company, regulators, police, animals and the local ecosystem. Rights to health, livelihood and peaceful protest compete with public order and economic continuity.

Immediate action should secure drinking water and medical relief, independently sample land, water and effluent, publish non-sensitive findings, and investigate both regulatory failure and any disproportionate policing. Closure should remain effective where serious non-compliance creates continuing harm, but an expert committee should assess whether a time-bound, monitored restart after proven treatment upgrades is environmentally safe. The precautionary principle governs uncertainty; polluter-pays places cleanup, health compensation and restoration costs on the company rather than taxpayers.

Workers need wage support, social-security access, retraining and temporary placement; ancillary firms need transition assistance, not immunity for the polluter. A local grievance forum should include residents, workers, technical experts and administration. Any restart must require zero-bypass monitoring, public compliance reports, escrow for remediation and automatic suspension triggers.

Vellore supports sustainable development, precaution and polluter-pays in environmental law, but the decision remains fact-specific. The ethical settlement therefore restores ecology and health, preserves viable livelihoods where possible, assigns costs to the wrongdoer and replaces reactive agitation with accountable regulation.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 2 - 2020 - 10 marks

Question: GS-IV Q5(b): "The current internet expansion has instilled a different set of cultural values which are often in conflict with traditional values." Discuss. (Answer in 150 words)

Source / ownership: Exact isolated Q5(b) verified against books\more_previous_papers\Gen_St_P4.pdf, page 3. Topic 13 owns the internet-culture subpart; Q5(a), gender inequality and Savitribai Phule, belongs to Topic 2.

Model solution

Internet expansion changes how identity, authority and community are formed. It can promote autonomy, scientific temper, gender voice, intercultural learning and scrutiny of oppressive customs. Young people can find support beyond restrictive local settings, and public institutions can communicate across language and distance.

Conflict arises when network values of speed, visibility, individual self-expression and global consumer culture meet traditions emphasising privacy, hierarchy, family authority and local continuity. The result may be generational tension, imitation, trolling, sexualised harassment, misinformation and polarisation. Yet neither category is morally pure: tradition can preserve solidarity and wisdom but also exclusion; digital culture can liberate but also manipulate attention and commodify intimacy.

The ethical response is critical adaptation, not blanket rejection or surrender. Digital literacy, consent, privacy, respectful dialogue, age-appropriate safeguards and platform accountability should filter harmful practices. Constitutional dignity and equality provide the standard: retain traditions compatible with rights, reform oppressive ones, and domesticate technology through humane civic values.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 3 - 2021 - 10 marks

Question: GS-IV Q2(a): Impact of digital technology as reliable source of input for rational decision making is a debatable issue. Critically evaluate with suitable example. (Answer in 150 words)

Source / ownership: Exact isolated Q2(a) verified against books\more_previous_papers\QP-CSM-21-GENSTUDIESPAPER-IV-110122.pdf, page 2. Topic 13 owns digital decision reliability; Q2(b), innovativeness in ethical dilemmas, is cross-linked to decision method.

Model solution

Digital technology can make decisions more rational by integrating large records, detecting patterns, applying consistent criteria and creating auditable timelines. For example, a district may use satellite and transaction data to prioritise field inspection of suspected benefit leakage, saving scarce staff time.

Reliability, however, is conditional. Incomplete records, biased samples, proxy variables, wrong objectives, cyber manipulation and model drift can produce precise-looking error. A digital score may omit disability, migration or local shocks that an experienced officer would recognise. Automation bias can then convert an advisory flag into an unreasoned denial, while opaque systems prevent citizens from correcting data.

Therefore digital input should be triangulated with verified records and field context. High-stakes use needs data-quality checks, explainable factors, security, human authority to depart, recorded reasons, appeal and periodic audit. Technology is a valuable epistemic aid, not an infallible moral agent. Rationality lies in reviewable evidence and accountable judgment, not in digitisation by itself.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 4 - 2022 - 10 marks

Question: GS-IV Q4(b): Online methodology is being used for day-to-day meetings, institutional approvals in the administration and for teaching and learning in education sector to the extent telemedicine in the health sector is getting popular with the approvals of the competent authority. No doubt, it has advantages and disadvantages for both the beneficiaries and the system at large. Describe and discuss the ethical issues involved in the use of online method particularly to the vulnerable section of the society. (Answer in 150 words)

Source / ownership: Exact isolated Q4(b) verified against books\more_previous_papers\QP-CSM-22-GENERAL-STUDIES-PAPER IV-190922.pdf, page 3. Topic 13 owns online-method ethics; Q4(a), good governance and e-governance initiatives, cross-links to Topic 11.

Model solution

Online methods improve reach, speed, continuity, documentation and reduced travel. Telemedicine can connect specialists to remote patients; online approvals can reduce delay; digital learning can widen access to material.

Their ethical value depends on substantive inclusion. Vulnerable persons may lack devices, connectivity, electricity, literacy, accessible design, private space or local-language support. Women may have less control over household devices; persons with disabilities may face inaccessible interfaces; elderly users may be exposed to fraud. Telemedicine also raises diagnostic limits, informed consent, confidentiality and data-security concerns. An online-only system can transfer administrative cost to the citizen and hide exclusion behind successful portal statistics.

A just design therefore combines assisted digital centres, accessibility standards, plain-language consent, minimal secure data, grievance redress and an effective offline alternative. High-risk health or entitlement decisions need human follow-up. Efficiency is legitimate, but equality, dignity and care require that no person loses an essential service merely because the channel is digital.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 5 - 2023 - 20 marks

Question: GS-IV Q12: You hold a responsible position in a ministry in the government. One day in the morning you received a call from the school of your 11-year-old son that you are required to come and meet the Principal. You proceed to the school and find your son in the Principal's office. The Principal informs you that your son had been found wandering aimlessly in the grounds during the time classes were in progress. The class teacher further informs you that your son has lately become a loner and did not respond to questions in the class; he had also been unable to perform well in the football trials held recently. You bring your son back from the school and in the evening, you along with your wife try to find out the reasons for your son's changed behaviour. After repeated cajoling, your son shares that some children had been making fun of him in the class as well as in the WhatsApp group of the students by calling him stunted, dumb and a frog. He tells you the names of a few children who are the main culprits but pleads with you to let the matter rest. After a few days, during a sporting event, where you and your wife have gone to watch your son play, one of your colleague's son shows you a video in which students have caricatured your son. Further, he also points out the perpetrators who were sitting in the stands. You purposefully walk past them with your son and go home. Next day, you find on social media a video denigrating you, your son and even your wife, stating that you engaged in physical bullying of children on the sports field. The video became viral on social media. Your friends and colleagues began calling you to find out the details. One of your juniors advised you to make a counter video giving the background and explaining that nothing had happened on the field. You, in turn, posted a video which you have captured during the sporting event, identifying the likely perpetrators who were responsible for your son's predicament. You have also narrated what has actually happened in the field and made attempts to bring out the adverse effects of the misuse of social media. (a) Based on the above case study, discuss the ethical issues involved in the use of social media. (b) Discuss the pros and cons of using social media by you to put across the facts to counter the fake propaganda against your family. (Answer in 250 words)

Source / ownership: Exact full case and both demands verified against books\more_previous_papers\QP-CSM-23-GENERAL-STUDIES-PAPER-IV-180923.pdf, pages 14-15. Topic 13 owns social-media ethics; Topic 22 supplies case architecture and child-sensitive implementation.

Model solution

The case involves child dignity and safety, cyberbullying, privacy, misinformation, parental responsibility, public-office restraint, due process and the permanence of digital identification. The original attackers exploited reach and anonymity; the false viral allegation harmed the whole family and public trust. Yet identifying likely minor perpetrators can expose children to retaliation before facts are independently established.

A counter-video offers speed, direct access to the same audience, evidence, correction and resistance to a one-sided narrative. It may reassure colleagues and expose platform misuse. Its disadvantages are escalation, context collapse, further circulation of the child's humiliation, privacy invasion, mistaken identification, vigilantism and the appearance that official status is being used in a private conflict.

I would prioritise the son's psychological safety, counselling and informed participation. I would preserve screenshots and originals, request the school to investigate under a child-protection and anti-bullying process, seek removal or correction through the platform, and use lawful cyber channels if threats or serious defamation persist. Any public response should be brief, factual and anonymised: deny physical bullying, state that evidence has been submitted to competent authorities, and avoid naming minors.

The already posted identifying video should be removed or edited, with a clarification protecting all children. Truthful counter-speech is legitimate, but proportionality and care demand verification, minimum disclosure and institutional remedy rather than a viral contest.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 6 - 2024 - 10 marks

Question: GS-IV Q1(a): The application of Artificial Intelligence as a dependable source of input for administrative rational decision-making is a debatable issue. Critically examine the statement from the ethical point of view. (Answer in 150 words)

Source / ownership: Exact isolated Q1(a) verified against books\mains\05 UPSC 2024 Paper-IV_Final 1.pdf, page 2. Topic 13 is primary owner; Q1(b), dimensions of ethics in professional decision-making, belongs to Topic 1.

Model solution

AI can be a dependable input when it detects patterns across large records, improves consistency, reduces routine delay and enables scarce officials to focus on difficult cases. Used to flag anomalies in welfare payments, it may strengthen stewardship of public funds.

Dependability is not inherent. Bias may enter data, design, deployment and feedback; generative systems may hallucinate; opaque scores weaken reason-giving; automation bias and de-skilling reduce practical wisdom. Accountability also diffuses among vendor, procurer and officer, although the affected citizen still bears the error. From a deontological view, dignity and due process require an intelligible, appealable decision. Consequentialism supports efficiency only after error and distributional harms are counted.

High-stakes administrative AI should therefore remain advisory. Representative testing, red-teaming, audit logs, plain-language explanations, data correction, meaningful human review, contestability and continuous monitoring are essential. The India AI Governance Guidelines provide policy guidance, not proof of fairness. AI is ethically dependable only as a verified and supervised input within an accountable human institution.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 7 - 2024 - 10 marks

Question: GS-IV Q2(b): Global warming and climate change are the outcomes of human greed in the name of development, indicating the direction in which extinction of organisms including human beings is heading towards loss of life on Earth. How do you put an end to this to protect life and bring equilibrium between the society and the environment? (Answer in 150 words)

Source / ownership: Exact isolated Q2(b) verified against books\mains\05 UPSC 2024 Paper-IV_Final 1.pdf, page 2. Topic 13 owns climate ethics; Q2(a), powerful nations and ongoing conflicts, belongs to Topic 12.

Model solution

The statement identifies greed as a vice: limitless consumption treats nature, poorer communities and future generations as instruments. Climate change is therefore not only a technical market failure but a failure of temperance, justice and stewardship. Its burdens are unequal: those contributing least often face the greatest heat, livelihood and disaster risks.

Equilibrium requires sustainable development: efficient and renewable energy, public transport, circular production, ecosystem restoration, climate-resilient agriculture and credible disclosure by firms. Polluter-pays should internalise damage, while precaution supports action despite uncertainty about exact local effects. Citizens must moderate wasteful consumption, but responsibility cannot be shifted from institutions to individual lifestyle alone.

Internationally, CBDR-RC combines common action with historical contribution and respective capability; it means differentiated, not absent, obligations for developing states. Participatory transitions and support for affected workers make mitigation just. The goal is not to end development, but to redefine it so present welfare no longer destroys the ecological conditions of future freedom and life.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 8 - 2024 - 20 marks

Question: GS-IV Q7: There is a technological company named ABC Incorporated which is the second largest worldwide, situated in the Third World. You are the Chief Executive Officer and the majority shareholder of this company. The fast technological improvements have raised worries among environmental activists, regulatory authorities, and the general public over the sustainability of this scenario. You confront substantial issues about the business's environmental footprint. In 2023, your organization had a significant increase of 48% in greenhouse gas emissions compared to the levels recorded in 2019. The significant rise in energy consumption is mainly due to the surging energy requirements of your data centers, fuelled by the exponential expansion of Artificial Intelligence (AI). AI-powered services need much more computational resources and electrical energy compared to conventional online activities, notwithstanding their notable gains. The technology's proliferation has led to a growing concern over the environmental repercussions, resulting in an increase in warnings. AI models, especially those used in extensive machine learning and data processing, exhibit much greater energy consumption than conventional computer tasks, with an exponential increase. Although there is already a commitment and goal to achieve net zero emissions by 2030, the challenge of lowering emissions seems overwhelming as the integration of AI continues to increase. To achieve this goal, substantial investments in renewable energy use would be necessary. The difficulty is exacerbated by the competitive environment of the technology sector, where rapid innovation is essential for preserving market standing and shareholders' worth. To achieve a balance between innovation, profitability and sustainability, a strategic move is necessary that is in line with both business objectives and ethical obligations. (a) What is your immediate response to the challenges posed in the above case? (b) Discuss the ethical issues involved in the above case. (c) Your company has been identified to be penalized by technological giants. What logical and ethical arguments will you put forth to convince about its necessity? (d) Being a conscience being, what measures would you adopt to maintain balance between AI innovation and environmental footprint? (Answer in 250 words)

Source / ownership: Exact full case and all four demands verified against books\mains\05 UPSC 2024 Paper-IV_Final 1.pdf, pages 4-5. Topic 13 owns AI-footprint and environmental substance; Topic 22 owns method and Topic 12 is a corporate-governance cross-link. The official case contains no algorithmic-discrimination fact.

Model solution

I would acknowledge the 48% increase, freeze avoidable high-carbon expansion pending a rapid independent footprint review, and protect essential services while the board adopts a verified transition plan. Stakeholders are users, employees, shareholders, regulators, energy and water-stressed communities, future generations and competitors.

The ethical issues are truthful disclosure versus greenwashing; profit and innovation versus climate duty; present shareholder value versus intergenerational equity; unequal burdens in a Third World setting; and responsibility for operational energy, cooling water, hardware and e-waste. A 2030 promise cannot excuse present inaction. Polluter-pays and sustainable development require costs to be internalised rather than shifted.

If stronger technological giants seek to penalise ABC, I would argue for transparent, uniform and capability-sensitive standards, independent verification and a fair transition period. Environmental accountability is necessary, but private rivals should not use it as discriminatory market exclusion. Equivalent footprints must face equivalent rules, with CBDR-RC used as an ethical analogy for capability, not as corporate immunity.

Measures include science-based interim carbon budgets; renewable procurement plus additional clean capacity; efficient models, hardware utilisation and workload scheduling; water-aware siting and cooling; repair, reuse and responsible e-waste; supplier standards; internal carbon pricing; audited public reporting; and executive incentives tied to absolute reductions. A board sustainability committee should monitor milestones and publish setbacks. Innovation should continue within resource budgets, with low-value computation deferred and high-social-value uses prioritised.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 9 - 2024 - 20 marks

Question: GS-IV Q12: Dr. Srinivasan is a senior scientist working for a reputed biotechnology company known for its cutting-edge research in pharmaceuticals. Dr. Srinivasan is heading a research team working on a new drug aimed at treating a rapidly spreading variant of a new viral infectious disease. The disease has been rapidly spreading across the world and the cases reported in the country are increasing. There is huge pressure on Dr. Srinivasan's team to expedite the trials for the drug as there is significant market for it, and the company wants to get the first-mover advantage in the market. During a team meeting, some senior team members suggest some shortcut for expediting the clinical trials for the drug and for getting the requisite approvals. These include manipulating data to exclude some negative outcomes and selectively reporting positive results, foregoing the process of informed consent and using compounds already patented by a rival company, rather than developing one's own component. Dr. Srinivasan is not comfortable taking such shortcuts, at the same time he realises meeting the targets is impossible without using these means. (a) What would you do in such a situation? (b) Examine your options and consequences in the light of the ethical questions involved. (c) How can data ethics and drug ethics save humanity at large in such a scenario? (Answer in 250 words)

Source / ownership: Exact full case and all three demands verified against books\mains\05 UPSC 2024 Paper-IV_Final 1.pdf, page 10. Topic 13 owns the data-integrity and consent cross-link; Topic 22 owns case method, with research and drug ethics supplying the specialist substantive route.

Model solution

I would refuse data manipulation, selective reporting, consent bypass and unauthorised use of patented compounds; preserve records; document the pressure; seek an urgent independent ethics and regulatory review; and propose a lawful accelerated protocol. Patients, trial participants, future users, researchers, the company, the rival patent holder, regulators and the wider public are stakeholders.

Compliance with the shortcuts may produce speed and first-mover gain, but it converts participants into means, conceals adverse effects, corrupts scientific knowledge, risks mass injury and destroys trust. Silent refusal protects personal integrity but leaves organisational pressure intact. Resignation may be necessary if correction fails, yet premature exit can abandon evidence. The strongest option is recorded internal escalation, independent review and competent external reporting where serious misconduct persists, with protection of confidential patient data.

Data ethics requires completeness, accuracy, reproducibility, secure handling, transparent adverse-result reporting and correction. Drug ethics requires scientific validity, informed and voluntary consent, favourable risk-benefit assessment, fair participant selection, monitoring and post-trial responsibility. Patent rights must be respected or addressed through lawful licensing and applicable public-health mechanisms, not covert appropriation.

A lawful accelerated design may use adaptive protocols, parallel administrative preparation and rapid independent review without weakening consent or evidence. These disciplines save humanity because unreliable data in a fast-moving epidemic scales harm globally. Urgency justifies faster ethical work, not lower truth, dignity or safety.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 10 - 2025 - 10 marks

Question: GS-IV Q1(a): In the present digital age, social media has revolutionised our way of communication and interaction. However, it has raised several ethical issues and challenges. Describe the key ethical dilemmas in this regard. (Answer in 150 words)

Source / ownership: Exact isolated Q1(a) verified against books\mains\UPSC Mains 2025 GS Paper 4.pdf, page 2. Topic 13 owns social-media ethics; Q1(b), constitutional morality and public service, belongs to Topic 14.

Model solution

Social media expands voice, association, emergency communication and accountability, but its ethical dilemmas arise from a business and design environment that rewards attention. Freedom of expression conflicts with protection from incitement, cyberbullying and defamation. Personalisation offers relevance but enables surveillance, micro-targeted manipulation and loss of autonomy. Virality democratises information yet accelerates misinformation, deepfakes and irreversible reputational harm before verification.

Further dilemmas include privacy versus public curiosity, anonymity versus answerability, platform moderation versus censorship, child safety versus participation, and global rules versus linguistic and cultural context. Algorithmic amplification can create polarisation and silence minorities even without direct state coercion.

An ethical response combines user digital literacy, consent and privacy by design, child-sensitive defaults, provenance and rapid correction tools, transparent moderation reasons, appeal, researcher access and proportionate lawful regulation. Government counter-speech should be factual and restrained. The aim is not a sanitised public sphere, but one where expression remains free while manipulation, hidden surveillance and preventable harm become contestable and accountable.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 11 - 2025 - 10 marks

Question: GS-IV Q2(b): Keeping the national security in mind, examine the ethical dilemmas related to controversies over environmental clearance of development projects in ecologically sensitive border areas in the country. (Answer in 150 words)

Source / ownership: Exact isolated Q2(b) verified against books\mains\UPSC Mains 2025 GS Paper 4.pdf, page 2. Topic 13 is the sole substantive owner of this subpart; Q2(a), Clausewitz and war as diplomacy, belongs to Topic 12.

Model solution

Border roads, communications and logistics can protect sovereignty, responders and remote communities. Yet ecologically sensitive mountains and forests may face irreversible habitat fragmentation, erosion, disaster risk and livelihood loss. The dilemma is therefore security and development versus precaution, community voice and intergenerational equity; neither value may be reduced to a slogan.

Blanket exemption risks using secrecy to avoid scrutiny. Complete public disclosure, however, may reveal genuine operational information. A proportionate process should define the specific security purpose, compare routes and technologies, choose the least ecologically damaging feasible alternative, consult affected communities on non-classified impacts, and subject classified material to cleared independent experts. Cumulative and disaster-risk assessment, seasonal construction controls, compensatory restoration, wildlife passages and continuous monitoring should follow.

Precaution does not ban all border infrastructure; it demands safeguards despite uncertainty. Reasons, conditions and compliance results should be public except for narrowly protected security details. Ethical clearance is thus a reviewable, least-damage accommodation of a legitimate security need, not automatic approval or automatic prohibition.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

Solved PYQ 12 - 2025 - 20 marks

Question: GS-IV Q8: In line with the Directive Principles of State Policy enshrined in the Indian Constitution, the government has a constitutional obligation to ensure basic needs - "Roti, Kapda aur Makan (Food, Clothes and Shelter)" - for the under-privileged. Pursuing this mandate, the district administration proposed clearing a portion of forest land to develop housing for the homeless and economically weaker sections of the society. The proposed land, however, is an ecologically sensitive zone densely populated with age-old trees, medicinal plants and vital biodiversity. Besides, these forests help to regulate micro-climate and rainfalls, provide habitat for wildlife, support soil fertility and prevent land/soil erosion and sustain livelihoods of tribal and nomadic communities. Inspite of the ecological and social costs, the administration argues in favour of the said proposal by highlighting that this very initiative addresses fundamental human rights as a critical welfare priority. Besides, it fulfils the government's duty to uplift and empower the poor through inclusive housing development. Further, these forest areas have become unsafe due to wild-animal threats and recurring human-wild life conflicts. Lastly, clearing forest-zones may help to curb anti-social elements allegedly using these areas as hideouts, thereby enhancing law and order. (a) Can deforestation be ethically justified in the pursuit of social welfare objectives like, housing for the homeless? (b) What are the socio-economic, administrative and ethical challenges in balancing environmental conservation with human development? (c) What substantial alternatives or policy interventions can be proposed to ensure that both environmental integrity and human dignity are protected? (Answer in 250 words)

Source / ownership: Exact full case and all three demands verified against books\mains\UPSC Mains 2025 GS Paper 4.pdf, pages 5-6. Topic 13 owns environmental substance; Topic 22 owns method. No degraded-forest, city-periphery or court-monitoring fact is added.

Model solution

Housing is a dignity and welfare imperative, but the stated forest is not vacant land. It contains age-old trees, medicinal plants, biodiversity, climate and rainfall regulation, wildlife habitat, fertile soil, erosion control and tribal and nomadic livelihoods. These ecological services support human rights across generations. Deforestation is therefore justifiable only after necessity and least-damaging alternatives are proved, not merely because the beneficiary goal is worthy.

Socio-economic challenges include land scarcity, affordability, livelihood displacement and human-wildlife conflict. Administrative challenges include reliable beneficiary selection, interdepartmental coordination, ecological valuation, consultation, rehabilitation and preventing capture. Ethical tensions oppose immediate shelter to intergenerational equity, anthropocentric welfare to biocentric and ecocentric value, and majority benefit to the voice of tribal and nomadic communities. Alleged hideouts do not establish that clearing is a necessary law-and-order response.

The administration should first inventory public land, vacant serviced plots and brownfield sites; use in-situ upgrading, rental housing, land pooling, vertical construction and distributed smaller sites; and improve transport to suitable land. It should map ecosystem services and wildlife corridors, consult affected communities, and compare lifecycle costs. If a residual diversion is demonstrably unavoidable, minimise its footprint, protect critical habitat and medicinal species, secure lawful livelihood rehabilitation, create wildlife mitigation and fund long-term restoration and monitoring.

The ethical verdict is integrated planning: fulfil shelter without treating the poor as a pretext for avoidable ecological loss or conservation as a reason to deny human dignity.

Why this earns marks: It states the governing ethical idea, translates it into public administration, uses a concrete Indian illustration, acknowledges a limit, and ends with a reasoned verdict.

ORIGINAL MAINS PRACTICE WITH MODEL SOLUTIONS

Original Mains Practice 1 - 10 marks

Question: Administrative AI should support judgment without becoming an unanswerable final decision-maker. Examine with suitable safeguards. Answer in about 150 words.

Model solution

Administrative AI can improve speed, consistency, anomaly detection and allocation of scarce inspection capacity. A welfare department may use it to flag unusual claims for human review. However, machine output reflects chosen data, labels, objectives and deployment conditions. It may reproduce exclusion, hallucinate, drift or omit context. Automation bias then makes a recommendation appear objective, while accountability is diffused among vendor, procurer and official.

The ethical position is decision support under human responsibility. Before procurement, the department should define purpose, rights impact and success criteria. It should require representative testing, security and bias red-teaming, documentation, audit access and an exit route. Deployment needs meaningful human authority to depart, intelligible reasons, notice of automation, correction of data, appeal and continuous monitoring. High-stakes decisions should never rest solely on unverified generative output.

The India AI Governance Guidelines offer useful policy guidance but do not prove a system fair or safe. AI becomes rational only inside a reviewable institution where an identified human remains answerable.

Why this earns marks: The response defines the issue, develops the relevant mechanism with India-centric evidence, tests a limitation, and gives a mark-scaled conclusion.

Original Mains Practice 2 - 10 marks

Question: Why are explainability, contestability and accountability distinct but mutually dependent safeguards in high-stakes AI? Answer in about 150 words.

Model solution

Explainability supplies intelligible reasons: the affected person should know the decisive factors, important limits and role of automation. Contestability converts knowledge into agency by allowing data correction, challenge, timely human reconsideration and remedy. Accountability assigns institutional responsibility and preserves logs, audits and sanctions so that failure has an answerable owner.

Each is insufficient alone. A technically accurate explanation without appeal merely describes an injustice. An appeal without reasons forces the citizen to guess what to challenge. Reasons and appeal without responsibility may end in the vendor and department blaming each other. In a scholarship-rejection system, the applicant should receive the relevant eligibility factors, correct an erroneous record and obtain a reasoned review by an authorised officer; the agency should retain audit logs and investigate recurring error.

Commercial secrecy may protect genuine intellectual property, but cannot erase due process. Risk-sensitive disclosure, independent confidential audit and plain-language adverse-action reasons can reconcile both interests. Together the three safeguards make automated power understandable, challengeable and answerable.

Why this earns marks: The response defines the issue, develops the relevant mechanism with India-centric evidence, tests a limitation, and gives a mark-scaled conclusion.

Original Mains Practice 3 - 15 marks

Question: A consent form alone does not make administrative data use ethical. Discuss through privacy, purpose limitation and the phased DPDP framework. Answer in about 200 words.

Model solution

Consent protects autonomy only when it is informed, specific, accessible and capable of withdrawal. In public administration, dependence on an essential service can make formal agreement weak: a citizen may click a bundled notice because refusal means losing a benefit. Ethical data use therefore also requires lawful authority, purpose limitation, minimisation, accuracy, retention limits, security and remedy.

Puttaswamy recognises privacy as a fundamental right linked to dignity and autonomy. A State restriction requires legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards. A useful scheme objective alone is insufficient. Thus a health database collected for treatment should not silently become a general profiling system.

The DPDP framework must be stated chronologically. G.S.R. 843(E) commenced specified provisions on 13 November 2025; the Consent Manager tranche begins 13 November 2026; core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027. It is wrong to call the Act fully operational at the present cut-off.

Nevertheless, deferred statutory commencement does not defer ethics. Administrators should use plain-language and granular notices, collect only necessary data, separate optional reuse, secure records, publish retention rules and provide correction and grievance channels. Consent is one part of a dignity-preserving architecture, not a universal moral waiver.

Why this earns marks: The response defines the issue, develops the relevant mechanism with India-centric evidence, tests a limitation, and gives a mark-scaled conclusion.

Original Mains Practice 4 - 15 marks

Question: Can the environmental precautionary principle guide governance of high-stakes AI? Critically examine the analogy and its limits. Answer in about 200 words.

Model solution

Precaution addresses decisions where scientific uncertainty coexists with threats of serious or irreversible harm. Rio Declaration Principle 15 gives its classic environmental formulation, and Vellore recognises precaution within Indian environmental law. High-stakes AI presents a structural analogy: model error may be opaque, scale rapidly and burden people who lack voice, while evidence of harm emerges only after deployment.

The analogy supports staged pilots, impact assessment, adversarial red-teaming, security tests, representative evaluation, fallback procedures, human oversight and continuous monitoring before mass use in welfare, policing or health. Lack of perfect certainty should not justify exposing citizens to an untested irreversible denial.

However, Rio and Vellore are environmental authorities, not binding Indian AI law. AI harms may be reversible through correction, while blanket precaution can freeze socially beneficial innovation and entrench incumbent vendors. Risk also varies: a translation assistant and an automated liberty decision cannot face identical controls.

Therefore precaution should operate as a proportionate ethical analogy. Classify impact, require stronger evidence where rights and irreversibility are greater, preserve contestability and review controls as performance changes. The principle justifies anticipatory governance, not a general technology veto or a false claim that Vellore directly regulates AI.

Why this earns marks: The response defines the issue, develops the relevant mechanism with India-centric evidence, tests a limitation, and gives a mark-scaled conclusion.

Original Mains Practice 5 - 20 marks

Question: Examine the full environmental footprint of AI through sustainable development and intergenerational justice. Suggest an accountable transition framework for India. Answer in about 250 words.

Model solution

AI's footprint is a lifecycle issue, not electricity alone. Training and repeated inference consume power; cooling and power generation consume water; semiconductor and server manufacture embody energy, minerals and pollution; rapid replacement produces electronic waste. Benefits and burdens are also spatially unequal: urban and global users may gain while water-stressed communities, workers near extraction and future generations bear costs.

Sustainable development requires meeting present needs without compromising future capacity. Intergenerational justice gives absent future citizens standing in today's resource allocation. Polluter-pays opposes shifting remediation to taxpayers, and environmental justice requires voice for affected communities. Yet a blanket restriction would ignore AI's public value in health, agriculture, disaster warning and accessible services.

An accountable Indian transition should begin with standardised disclosure of operational energy, water, location, embodied hardware and e-waste, without unsupported precision where measurement is immature. Public procurement can reward efficient models, repairability, renewable additionality, water-aware siting and responsible end-of-life handling. Operators should use carbon and water budgets, efficient architectures, high utilisation, workload scheduling, heat reuse where feasible and verified supplier standards. EIA and local consultation should apply where the governing law requires them; environmental claims need independent assurance.

Policy should prioritise high-social-value computation, support clean grids and domestic circular hardware capacity, and protect workers and communities during transition. Periodic public review must compare absolute impacts, not efficiency claims alone. The ethical goal is useful AI within ecological budgets, not innovation whose hidden costs narrow future freedom.

Why this earns marks: The response defines the issue, develops the relevant mechanism with India-centric evidence, tests a limitation, and gives a mark-scaled conclusion.

Original Mains Practice 6 - 20 marks

Question: A State proposes an AI system to select welfare beneficiaries and a water-intensive data centre to operate it in a drought-prone district. As district head, resolve the combined data, justice and environmental dilemma. Answer in about 250 words.

Model solution

Stakeholders include eligible applicants, excluded families, local residents and farmers, tribal or marginalised groups, officials, the vendor, data-centre workers, the power and water utilities, the ecosystem and future citizens. The proposal promises faster targeting and jobs, but creates linked risks: biased exclusion, opaque reasons, privacy loss, vendor dependence, groundwater stress, energy emissions and unequal distribution of benefits and burdens.

I would not approve immediate full-scale deployment. First, separate and test the two decisions. For beneficiary selection, define lawful eligibility, prohibit the model from making final adverse decisions, audit data and proxies, red-team vulnerable cases, provide notice, plain-language reasons, data correction, assisted offline application and timely human appeal. Contracts must assign accountability, preserve logs, permit independent audit and allow exit. Puttaswamy proportionality and purpose limitation govern personal-data use.

For the data centre, commission a transparent assessment of water source, drought scenarios, energy, cooling, hardware and e-waste; compare less water-intensive locations and architectures; consult affected communities; and select the least-damaging feasible alternative. Approval, if justified, should impose water and carbon budgets, renewable additionality, non-potable or recycled water where safe, public monitoring, drought curtailment triggers and restoration security. Claimed welfare benefits cannot override ecological limits.

A limited pilot using existing lower-impact infrastructure may proceed if independent review confirms rights and reliability safeguards. The final decision should publish reasons and residual risks. Innovation is acceptable only when beneficiaries can contest digital power and host communities do not subsidise it with dignity, water and future resilience.

Why this earns marks: The response defines the issue, develops the relevant mechanism with India-centric evidence, tests a limitation, and gives a mark-scaled conclusion.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

> **Subject:** Ethics | **Tier:** Advanced | **GS Paper:** GS-IV. **Core area:** Deeper treatment of AI-governance ethics, data ethics and environmental/climate ethics, with disciplined moral-theory application. **Grounded in:** Official UPSC GS-IV syllabus framing; audited GS-IV PYQs (2024-2025 Mains); standard AI-governance and environmental-ethics literature (precautionary principle, CBDR). FACT: = source/PYQ-grounded | ANALYSIS: = inference/analysis | CURRENT: = current anchor. *Companion:* `basic/13_Emerging-Ethics-Technology-AI-and-Environment.md`.

1. Architecture

```
AI-ETHICS RISK LAYER                   ENVIRONMENT/CLIMATE-ETHICS RISK LAYER
Bias in training data                    Discounting future/dispersed harm
Opacity ("black box" reasoning)          Scientific uncertainty exploited to delay action
Accountability diffusion                 Displacement of environmental cost onto
(who is responsible for an                vulnerable/voiceless groups (future
 automated decision?)                      generations, downstream communities)
De-skilling/over-reliance                 National-security vs ecological-protection
 (human judgment atrophies)                trade-offs in sensitive/border areas
        \                                      /
         \                                    /
          v                                  v
        COMMON CORRECTIVE PRINCIPLE:
   precaution + transparency + human/institutional accountability
   BEFORE scale-up, not retrofitted after harm is discovered
```

Analytical claim: AI-ethics and environment-ethics dilemmas share a **common structural risk** - decision-makers are tempted to defer precaution because harm is diffuse, delayed, or borne by those without a voice in the decision (algorithmically affected citizens; future generations; border-area ecological communities) - and GS-IV rewards candidates who name this shared structure explicitly rather than treating AI and environment as unrelated topics.

2. Concepts and distinctions

Concept	Precise meaning
ANALYSIS: **Algorithmic bias**	Systematic, unfair skew in an AI system's outputs arising from biased training data, biased feature selection, or biased labelling - administratively dangerous because it can appear "objective" while encoding historical discrimination.
ANALYSIS: **Explainability/interpretability**	The degree to which an AI system's decision process can be understood and audited by humans - a precondition for accountability in high-stakes administrative use (welfare eligibility, law-enforcement risk scoring).
FACT: **Precautionary principle - precise formulation**	"Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation" - **Rio Declaration, Principle 15 (1992)**, which frames it as a "precautionary **approach**", "widely applied by States according to their capabilities". Precaution can inform high-stakes AI governance **by analogy**; Rio Principle 15 and *Vellore* are environmental-law authorities, not a self-executing Indian AI rule.
FACT: **CBDR-RC** (Common but Differentiated Responsibilities and Respective Capabilities)	UNFCCC Article 3.1 requires Parties to protect the climate system "on the basis of equity and in accordance with their common but differentiated responsibilities and respective capabilities"; Article 4.1 applies it to all Parties' commitments; Rio Principle 7 is its source formulation; the Paris Agreement Article 2(2) adds "in the light of different national circumstances". It is the basis for differentiated climate-finance and mitigation expectations.
ANALYSIS: **Intergenerational equity**	The principle (echoed in India's own environmental jurisprudence, e.g., the public-trust doctrine) that the present generation holds natural resources in trust for future generations, not as exhaustible private property.

3. Detailed treatment

AI in administrative decision-making - the 2024 GS-IV question decomposed

- FACT: 2024 GS-IV Q1(a): "The application of Artificial Intelligence as a dependable source of input for administrative rational decision-making is a debatable issue."
- ANALYSIS: Advanced decomposition using the moral-theory toolkit (`08`): **deontological** concern - does algorithmic decision-making respect due process and the right to a reasoned, appealable decision, or does it reduce the citizen to a data point without dignity/voice? **Consequentialist** concern - does AI improve aggregate decision accuracy/speed/consistency (a real efficiency gain) enough to outweigh bias/error risk? **Virtue-ethics** concern - does reliance on AI erode the practical wisdom (phronesis) of the human official over time (de-skilling), weakening judgment exactly when a genuinely novel case falls outside the algorithm's training distribution?
- ANALYSIS: Strong answer structure: AI can be a *useful input*, not a "dependable source" in the sense of a final, unreviewed decision-maker - recommend mandatory human-in-the-loop review for consequential decisions, published/auditable decision criteria, and periodic bias audits.

Climate change as an ethics-of-consumption question - the 2024 GS-IV framing

- FACT: 2024 GS-IV Q2(b): frames global warming/climate change explicitly as "the outcomes of human greed in the name of development," warning of extinction of organisms "including human beings."
- ANALYSIS: Advanced reading: this framing invites a **virtue-ethics critique of consumption** (greed as a vice, temperance as the corrective virtue) *combined with* a **consequentialist/intergenerational** argument (present consumption imposing catastrophic costs on future, voiceless generations) - a strong answer names both registers rather than treating climate ethics as purely a policy- technical (emissions-target) question.
- ANALYSIS: India-specific balancing: CBDR-RC requires India (a still-developing, historically low per-

capita emitter) to accept differentiated but real responsibility - the ethical answer is neither "developing countries owe nothing" nor "all countries owe identical action," but proportionate, capability-linked responsibility.

Environmental clearance in ecologically sensitive border areas - the 2025 GS-IV dilemma

- FACT: 2025 GS-IV Q2(b): examines "ethical dilemmas related to controversies over environmental clearance of development projects in ecologically sensitive border areas," explicitly "keeping the national security in mind."
- ANALYSIS: Advanced decomposition: this is a genuine multi-value dilemma (not a simple pro-environment answer) - national-security infrastructure (roads, communication, forward posts) in border regions serves a constitutionally legitimate state interest, while ecologically sensitive zones carry precautionary-principle protection needs and often overlap with indigenous/local community rights.
- ANALYSIS: A defensible resolution: apply proportionality (minimum ecological footprint necessary for the specific security objective), require rigorous, transparent (though appropriately classified where genuinely security-sensitive) environmental impact assessment, and build in post-project ecological monitoring/restoration obligations - rather than either blanket exemption for "national security" projects or blanket prohibition on development in sensitive zones.

4. Institutional and reform architecture

- CURRENT: India's Digital Personal Data Protection Act, 2023 and the **India AI Governance Guidelines (MeitY, 5 November 2025)** are distinct domestic layers: DPDP governs personal-data processing, while the Guidelines articulate seven "sutras" (trust as the foundation; people first; innovation over restraint; fairness and equity; accountability; understandable by design; safety, resilience and sustainability) and six pillars for AI. They do not amount to one comprehensive AI statute - cross-link `Science-and-Technology/basic/09_Artificial-Intelligence-Governance-and-IndiaAI.md` and `Science-and-Technology/basic/12_Data-Protection-DPDP-Act-and-Cybersecurity.md` for statutory detail, to avoid duplicating that policy content here.
- CURRENT: Internationally, UNESCO's *Recommendation on the Ethics of Artificial Intelligence* (23 November 2021) and the OECD AI Principles (2019, updated May 2024) supply the standard, citable normative vocabulary - proportionality and do-no-harm, human oversight, transparency and explainability, accountability, fairness and non-discrimination, sustainability.
- FACT: India's environmental-clearance regime (EIA Notification, National Green Tribunal) is the domestic institutional mechanism operationalising the precautionary principle - see `Environment-and-Ecology/basic/16_Environmental-Impact-Assessment-and-NGT.md` for procedural detail.
- FACT: CBDR-RC is operationalised internationally through differentiated Nationally Determined Contributions (NDCs) under the Paris Agreement and climate-finance transfer mechanisms - see `Environment-and-Ecology/basic/19_UNFCCC-COP-Kyoto-Paris-Agreement.md` for treaty-law detail.

5. Indian applications and boundary cases

- ANALYSIS: A state welfare department using an AI tool to flag potentially fraudulent beneficiary claims must retain a human appeal mechanism and publish the flagging criteria, to avoid the "algorithmic accountability diffusion" risk identified above.
- ANALYSIS: Boundary case: a border-area hydropower or road project justified purely on national-security grounds, without transparent, published environmental impact assessment, risks converting a legitimate security exception into a precedent for bypassing precaution altogether - the ethical safeguard is a case-specific, proportionality-tested exception, not a categorical carve-out.

6. Limitations and trade-offs

- ANALYSIS: Mandatory human-in-the-loop review for all AI-assisted decisions can slow down high-volume administrative processes (e.g., large-scale welfare screening), creating a genuine efficiency- accountability trade-off that must be calibrated by the decision's stakes (irreversibility, fundamental-rights impact), not applied uniformly to every automated process.
- ANALYSIS: CBDR-RC, while ethically defensible, faces the practical difficulty that current and future emissions trajectories (not only historical ones) increasingly determine climate outcomes, creating tension between historical-responsibility and forward-looking-capability framings of fairness.
- ANALYSIS: Precaution applied too broadly in border-security contexts risks being invoked to indefinitely delay legitimate security infrastructure; the proportionality test is essential to prevent precaution itself from becoming a pretext for inaction.

7. Must-Know Facts for Advanced Prelims

- FACT: The precautionary principle's classical formulation is Principle 15 of the 1992 Rio Declaration on Environment and Development, which uses the term "precautionary approach".
- FACT: CBDR appears in Rio Principle 7 and UNFCCC Articles 3.1 and 4.1 (1992); the Paris Agreement (Article 2(2), 2015) states CBDR-RC "in the light of different national circumstances" and underpins the differentiated NDC structure.
- ANALYSIS: "Algorithmic accountability" and "explainability" are standard, widely used terms in current global AI-governance discourse (UNESCO's 2021 Recommendation, the OECD AI Principles, EU AI Act debates), not India-specific coinages.

8. Advanced Prelims traps

- WRONG: CBDR means developing countries have no climate obligations at all. -> It means differentiated, capability-linked obligations, not zero obligation - developing countries, including India, still commit to Nationally Determined Contributions under the Paris Agreement.
- WRONG: The precautionary principle requires certainty of harm before acting. -> It explicitly applies precisely *because* full scientific certainty is absent, for serious/irreversible risks.
- WRONG: AI bias is primarily a technical (coding) problem with no ethical dimension. -> Bias typically originates in socially/historically skewed training data and reflects real-world discrimination, making it fundamentally an ethics-of-fairness problem, not merely a technical bug.

9. CURRENT: Current-anchor application

Verified current anchor	Topic-specific analytical use
CURRENT: India AI Governance Guidelines (MeitY, 5 November 2025) - seven sutras and six pillars	Provide the dated domestic principles anchor for human-centric, fair, accountable, understandable, safe and sustainable AI; distinguish this non-statutory framework from DPDP law.
CURRENT: India AI Impact Summit, Bharat Mandapam, New Delhi, 19-20 February 2026 (within a 16-20 February programme)	Its *Sarvajana Hitaya, Sarvajana Sukhaya* framing, and the People-Planet-Progress impact framing, supply a current bridge between inclusion, human agency and AI's environmental footprint.
CURRENT: UNESCO Recommendation on the Ethics of AI (23 November 2021); OECD AI Principles (2019, updated May 2024)	Give internationally agreed normative vocabulary for human oversight, explainability and accountability that an Indian answer can cite without over-claiming domestic statutory force.
CURRENT: Paris Agreement NDC cycles (ongoing, periodically updated)	Anchors CBDR-RC discussion in the current international climate-governance architecture rather than only the 1992 UNFCCC baseline.

10. PYQ-based analytical application

- FACT: 2024 GS-IV Q1(a) rewards naming specific AI risks (bias, opacity, accountability diffusion, de-skilling) and a specific institutional safeguard (human-in-the-loop, audit, explainability requirement) rather than a generic "AI is a double-edged sword" answer.
- FACT: 2025 GS-IV Q2(b) rewards explicit proportionality-based resolution of the national-security-vs-ecology tension, naming the precautionary principle, rather than a one-sided answer favouring either development or conservation exclusively.

11. Mains-ready framework

Central thesis: Technological and environmental ethics dilemmas share a structural risk of deferred precaution against diffuse, delayed or voiceless harm; the ethical response is to build in accountability, transparency and proportionality *before* scale-up, using human-in-the-loop review, precautionary assessment and capability-linked responsibility-sharing.

1. Name the specific risk (bias/opacity/accountability diffusion for AI; irreversible/diffuse harm for environment/climate).
2. Apply the precautionary principle and/or CBDR-RC by name where relevant.
3. Triangulate using deontology (rights/dignity floor), consequentialism (aggregate welfare, long time-horizon) and virtue ethics (temperance, practical wisdom, avoiding de-skilling).
4. Recommend a specific, proportionate institutional safeguard (human review, audit, transparent EIA, capability-linked target).
5. Acknowledge the trade-off/limitation of your own recommendation.

12. Probable questions

- ANALYSIS: **Prelims:** State the precautionary principle's classical formulation and its source document.
- ANALYSIS: **Mains (10 marks):** Why can AI be described as a "debatable" input for administrative decision-making? Suggest specific safeguards.
- ANALYSIS: **Mains (15 marks):** Examine the ethical dilemmas in environmental clearance for development projects in ecologically sensitive border areas, balancing national security, ecological protection and community rights.

13. Study links

- FACT: Foundation companion: `basic/13_Emerging-Ethics-Technology-AI-and-Environment.md`.
- FACT: `12_Corporate-Governance-and-International-Ethics.md` - the corporate AI/environment case study.
- FACT: `Science-and-Technology/basic/09_Artificial-Intelligence-Governance-and-IndiaAI.md` - AI policy detail.
- FACT: `Environment-and-Ecology/basic/16_Environmental-Impact-Assessment-and-NGT.md` and

`Environment-and-Ecology/basic/19_UNFCCC-COP-Kyoto-Paris-Agreement.md` - full statutory/treaty depth.

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md`.

- **Years represented:** 2018, 2020, 2021, 2022, 2023
- **Paper(s):** GS-IV
- **Routed question demands:** 7

Semantic table split - Year

- **Year:** 2018
- **Paper:** GS-IV
- **Q:** 5
- **PYQ demand (neutral rendering):** (a) government dam construction in forest valley inhabited by ethnic communities - rational policy for unforeseen contingencies; (b) process of resolving ethical dilemmas in public administration
- **Directive / format:** Suggest/Explain · 10 + 10 marks · 150 words each
- **Source status:** Routed to owning Ethics topic
- **Owner requirement:** Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
- **Year:** 2018
- **Paper:** GS-IV
- **Q:** 10
- **PYQ demand (neutral rendering):** Corporate chemical factory ordered closed after pollution protests and public agitation; closure causes unemployment in workers and ancillary units; address as senior officer
- **Directive / format:** Case study · 20 marks · 250 words
- **Source status:** Case routed to Ethics case-study method
- **Owner requirement:** Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
- **Year:** 2020
- **Paper:** GS-IV
- **Q:** 5
- **PYQ demand (neutral rendering):** (a) main factors for gender inequality in India and Savitribai Phule contribution; (b) internet expansion instilling cultural values in conflict with traditional values
- **Directive / format:** Discuss · 10 + 10 marks · 150 words each
- **Source status:** Routed to owning Ethics topic
- **Owner requirement:** Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

- **Year:** 2021
- **Paper:** GS-IV
- **Q:** 2
- **PYQ demand (neutral rendering):** (a) digital technology as reliable input for rational decision making - critically evaluate; (b) innovativeness and creativity in resolving ethical dilemmas
- **Directive / format:** Critically evaluate / Discuss · 10 + 10 marks · 150 words each
- **Source status:** Routed to owning Ethics topic
- **Owner requirement:** Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
- **Year:** 2022
- **Paper:** GS-IV
- **Q:** 4
- **PYQ demand (neutral rendering):** (a) good governance and e-governance initiatives for beneficiaries; (b) ethical issues in online methodology affecting vulnerable sections of society
- **Directive / format:** Discuss · 10 + 10 marks · 150 words each
- **Source status:** Routed to owning Ethics topic
- **Owner requirement:** Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
- **Year:** 2022
- **Paper:** GS-IV
- **Q:** 12
- **PYQ demand (neutral rendering):** Case on Environment Pollution Control Board officer enforcing industrial environmental compliance despite resistance and threats
- **Directive / format:** Case study · 20 marks · 250 words
- **Source status:** Case routed to Ethics case-study method
- **Owner requirement:** Apply stakeholders, dilemmas, options, justification, implementation and safeguards.
- **Year:** 2023
- **Paper:** GS-IV
- **Q:** 12
- **PYQ demand (neutral rendering):** Case on senior government official managing son's cyberbullying and posting a counter-video on social media
- **Directive / format:** Case study · 20 marks · 250 words
- **Source status:** Case routed to Ethics case-study method
- **Owner requirement:** Apply stakeholders, dilemmas, options, justification, implementation and safeguards.

What this owner must now support

- (a) government dam construction in forest valley inhabited by ethnic communities - rational policy for unforeseen contingencies; (b) process of resolving ethical dilemmas in public administration
- Corporate chemical factory ordered closed after pollution protests and public agitation; closure causes unemployment in workers and ancillary units; address as senior officer
- (a) main factors for gender inequality in India and Savitribai Phule contribution; (b) internet expansion instilling cultural values in conflict with traditional values
- (a) digital technology as reliable input for rational decision making - critically evaluate; (b) innovativeness and creativity in resolving ethical dilemmas
- (a) good governance and e-governance initiatives for beneficiaries; (b) ethical issues in online methodology affecting vulnerable sections of society

- Case on Environment Pollution Control Board officer enforcing industrial environmental compliance despite resistance and threats
- Case on senior government official managing son's cyberbullying and posting a counter-video on social media

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

CONSOLIDATED REGISTER NOTES

EMBEDDED TWELVE-PANEL ASCII REVISION ATLAS

The panels compress the learning route without replacing the complete teaching and solved practice above.

ASCII PANEL 1/12 - 1. AI administrative decision chain

```
CENTRAL FOCUS
1. AI administrative decision chain
  [1] Define lawful public purpose
        |
        v
  [2] Check data fitness
        |
        v
  [3] Generate advisory output
        |
        v
  [4] Show decisive factors
        |
        v
  [5] Apply human judgment
        |
        v
  [6] Record independent reasons
        |
        v
  [7] Offer appeal and correction
        |
        v
  [8] Audit outcomes and drift
        |
        v
VERDICT -> AI may inform power; an answerable institution must exercise it.
ANSWER USE -> Use for 2021 Q2(a) and 2024 Q1(a).
```

ASCII PANEL 2/12 - 2. Bias control across the lifecycle

```
CENTRAL FOCUS
2. Bias control across the lifecycle
  [1] Test representation in data
        |
        v
  [2] Review labels and proxies
        |
        v
  [3] Question design objective
        |
        v
  [4] Red-team unequal impact
        |
        v
  [5] Check deployment conditions
        |
        v
  [6] Monitor subgroup outcomes
        |
        v
  [7] Break feedback loops
        |
        v
  [8] Correct and retrain safely
        |
        v
VERDICT -> Bias is a socio-technical lifecycle risk, not merely a coding defect.
ANSWER USE -> Use to replace generic claims that AI is biased.
```

ASCII PANEL 3/12 - 3. Contestability and remedy

```
CENTRAL FOCUS
3. Contestability and remedy
  [1] Notify material automation
        |
        v
  [2] Give intelligible reasons
        |
        v
  [3] Let citizen inspect data
        |
        v
  [4] Enable factual correction
        |
        v
  [5] Provide simple challenge
        |
        v
  [6] Require human reconsideration
        |
        v
  [7] Order correction or remedy
        |
        v
  [8] Feed lessons into redesign
        |
        v
VERDICT -> A right without a usable challenge route leaves automated power unchecked.
ANSWER USE -> Use for explainability, due process and accountability.
```

ASCII PANEL 4/12 - 4. Privacy governance chain

```
CENTRAL FOCUS
4. Privacy governance chain
  [1] State specific purpose
        |
        v
  [2] Establish lawful basis
        |
        v
  [3] Seek meaningful consent
        |
        v
  [4] Collect minimum data
        |
        v
  [5] Secure and limit access
        |
        v
  [6] Restrict reuse and retention
        |
        v
  [7] Enable correction and exit
        |
        v
  [8] Remedy breach and misuse
        |
        v
VERDICT -> Privacy protects dignity and autonomy throughout the data lifecycle.
ANSWER USE -> Use for consent, purpose and minimisation answers.
```

ASCII PANEL 5/12 - 5. Puttaswamy proportionality test

```
CENTRAL FOCUS
5. Puttaswamy proportionality test
  [1] Privacy is a fundamental right
        |
        v
  [2] Identify legal authority
        |
        v
  [3] Name legitimate State aim
        |
        v
  [4] Prove rational connection
        |
        v
  [5] Test least-restrictive means
        |
        v
  [6] Balance rights and benefit
        |
        v
  [7] Add procedural safeguards
        |
        v
  [8] Keep review and remedy open
        |
        v
VERDICT -> A useful aim does not by itself justify a privacy restriction.
ANSWER USE -> Use as the constitutional spine for State data use.
```

ASCII PANEL 6/12 - 6. Social-media ethics map

```
CENTRAL FOCUS
6. Social-media ethics map
  [1] Expression and association
        |
        v
  [2] Misinformation and deepfakes
        |
        v
  [3] Manipulation and addiction
        |
        v
  [4] Privacy and surveillance
        |
        v
  [5] Cyberbullying and dignity
        |
        v
  [6] Polarisation and amplification
        |
        v
  [7] Moderation reasons and appeal
        |
        v
  [8] Verification and proportionate law
        |
        v
VERDICT -> Protect open debate while making hidden manipulation and preventable harm
             answerable.
ANSWER USE -> Use for 2023 Q12 and 2025 Q1(a).
```

ASCII PANEL 7/12 - 7. AI environmental footprint

```
CENTRAL FOCUS
7. AI environmental footprint
  [1] Training electricity demand
        |
        v
  [2] Repeated inference demand
        |
        v
  [3] Cooling and power water
        |
        v
  [4] Hardware embodied impacts
        |
        v
  [5] Mineral and supply-chain harm
        |
        v
  [6] Replacement and e-waste
        |
        v
  [7] Absolute-impact disclosure
        |
        v
  [8] Budgets, efficiency and repair
        |
        v
VERDICT -> A renewable-power claim alone does not establish low-impact AI.
ANSWER USE -> Use for 2024 Q7 and lifecycle analysis.
```

ASCII PANEL 8/12 - 8. Precaution before scale-up

```
CENTRAL FOCUS
8. Precaution before scale-up
  [1] Identify serious possible harm
        |
        v
  [2] Map uncertainty and exposure
        |
        v
  [3] Test reversible pilot
        |
        v
  [4] Compare safer alternatives
        |
        v
  [5] Apply proportionate controls
        |
        v
  [6] Set stop and fallback rules
        |
        v
  [7] Monitor emerging evidence
        |
        v
  [8] Revise before wider deployment
        |
        v
VERDICT -> Rio and Vellore guide AI only by analogy, not as binding AI law.
ANSWER USE -> Use for anticipatory AI and environmental governance.
```

ASCII PANEL 9/12 - 9. Environmental justice test

```
CENTRAL FOCUS
9. Environmental justice test
  [1] Identify ecological benefits
        |
        v
  [2] Identify pollution and loss
        |
        v
  [3] Map burdened communities
        |
        v
  [4] Include future generations
        |
        v
  [5] Secure meaningful participation
        |
        v
  [6] Compare distributional options
        |
        v
  [7] Assign compensation and cleanup
        |
        v
  [8] Provide accessible remedy
        |
        v
VERDICT -> A green or welfare label cannot hide unequal burdens and silenced voices.
ANSWER USE -> Use for pollution, housing and siting cases.
```

ASCII PANEL 10/12 - 10. Sustainable-development bridge

```
CENTRAL FOCUS
10. Sustainable-development bridge
  [1] Define present human need
        |
        v
  [2] Value ecosystem services
        |
        v
  [3] Protect future capability
        |
        v
  [4] Compare lower-impact routes
        |
        v
  [5] Internalise pollution costs
        |
        v
  [6] Support a just transition
        |
        v
  [7] Monitor real outcomes
        |
        v
  [8] Revise if limits are breached
        |
        v
VERDICT -> Development and ecology are jointly constrained, not automatic overrides.
ANSWER USE -> Use for climate ethics and 2025 Q8.
```

ASCII PANEL 11/12 - 11. Border environmental clearance

```
CENTRAL FOCUS
11. Border environmental clearance
  [1] Specify security objective
        |
        v
  [2] Map sensitive ecology
        |
        v
  [3] Compare route alternatives
        |
        v
  [4] Use least-damaging feasible design
        |
        v
  [5] Consult on non-classified impacts
        |
        v
  [6] Review secrets through cleared experts
        |
        v
  [7] Condition restoration and monitoring
        |
        v
  [8] Publish reasons within security limits
        |
        v
VERDICT -> Security warrants tailored protection, not automatic ecological exemption.
ANSWER USE -> Use for 2025 Q2(b).
```

ASCII PANEL 12/12 - 12. Climate fairness architecture

```
CENTRAL FOCUS
12. Climate fairness architecture
  [1] Common duty to protect climate
        |
        v
  [2] Historical contribution matters
        |
        v
  [3] Respective capability matters
        |
        v
  [4] Current emissions still matter
        |
        v
  [5] CBDR-RC differentiates effort
        |
        v
  [6] Finance and technology enable action
        |
        v
  [7] Protect vulnerable transitions
        |
        v
  [8] Retain reviewable national duties
        |
        v
VERDICT -> Differentiated responsibility means fair action, not zero obligation.
ANSWER USE -> Use for 2024 Q2(b) and climate justice.
```

ONE-PAGE CONCEPT GRID

Concept / thinker	Exam-ready formulation
AI is decision support, not an accountability substitute	Administrative AI can improve speed, consistency and pattern detection, but it should remain a decision-support input whose recommendation is tested against law, context and reasons by an answerable public official.
Bias can enter data, design, deployment and feedback	AI unfairness may originate in unrepresentative data, labels and proxy variables, design objectives, unequal deployment conditions, or feedback loops that turn earlier skewed outputs into future training evidence.
Explainability must be useful to the affected person	Explainability in high-stakes administration requires an intelligible account of the decisive factors, limits and review route, not merely disclosure of source code or a technical statement that only specialists can understand.
Human oversight must be meaningful	Human oversight is meaningful only when the reviewer has competence, time, authority, relevant information and freedom to depart from an automated recommendation; ceremonial approval preserves neither judgment nor accountability.
Accountability requires an auditable responsibility chain	Algorithmic accountability assigns named responsibility across procurer, developer, deploying department and final decision-maker, while logs, impact assessment, audit trails and incident reporting make conduct reviewable before and after harm.
Contestability gives affected persons an effective remedy	Contestability means a person can know that automation materially influenced a decision, challenge relevant data and reasoning, obtain timely human reconsideration, and secure correction or remedy without prohibitive cost.
Automation bias and de-skilling are institutional risks	Automation bias is the tendency to over-trust machine output, while de-skilling is the gradual erosion of human expertise through disuse; both require calibrated reliance, training and periodic unaided judgment.
Hallucination and reliability require verification	Generative AI may produce fluent but false citations, facts or explanations, so reliability demands source verification, uncertainty disclosure, domain testing and prohibition of unverified output as the sole basis of consequential decisions.
Procurement must include red-teaming and monitoring	Responsible AI procurement specifies the public purpose, representative testing, security and bias red-teaming, documentation, audit access, change control, incident response, continuing performance monitoring and an exit route.
Deepfakes attack authenticity and public trust	Deepfakes create an authenticity problem by making fabricated audio or video appear evidentially real; proportionate responses combine provenance signals, verification, rapid correction, platform process and due regard for lawful expression.
Social media combines misinformation, manipulation and privacy harms	Social media dilemmas include falsehood, micro-targeted manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation; speed and reach magnify harm, while blanket censorship can damage expression and democratic debate.
The digital divide is about capability, not connectivity alone	Digital inclusion requires affordable access, devices, language and disability accessibility, literacy, assistance and an effective offline alternative; an online portal can otherwise convert administrative efficiency into substantive exclusion.

CORE REVISION SPINE

- **Privacy protects dignity, autonomy and contextual control:** Privacy is not secrecy alone: it protects dignity, decisional autonomy and a person's reasonable control over how information given in one context is collected, combined, inferred and used in another.
- **Puttaswamy requires the full proportionality inquiry:** A State privacy restriction should be tested for legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards; a useful public objective alone does not complete the inquiry.
- **DPDP commencement is phased, not fully operational:** G.S.R. 843(E) commenced a specified immediate tranche on 13 November 2025; the Consent Manager tranche begins 13 November 2026, while core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027.
- **Consent must be purpose-bound and security-backed:** Ethical data processing joins informed and revocable consent with purpose limitation, data minimisation, accuracy, retention limits and reasonable security; a clicked box cannot authorise indefinite collection or unrelated reuse.
- **AI has energy, water, hardware and e-waste costs:** AI's environmental footprint spans electricity for training and inference, water used in cooling and power generation, embodied impacts of hardware manufacture, mineral extraction, replacement cycles and electronic waste.
- **Sustainable development is a two-generation bridge:** Sustainable development meets present needs without compromising future generations' ability to meet their own needs; it neither makes development absolute nor converts environmental protection into an automatic veto.
- **Intergenerational equity represents absent future citizens:** Intergenerational equity extends public interest across time by requiring present decision-makers to account for durable resource depletion, climate risk and ecological damage imposed on people unable to participate in today's choice.
- **Precaution may guide high-stakes AI only by analogy:** Rio Principle 15 and Vellore are environmental authorities; their precaution logic may inform high-stakes AI by analogy where uncertain harm could be serious, but they are not binding Indian AI law.
- **Polluter-pays allocates prevention and remediation costs:** Polluter-pays requires the actor causing environmental harm to bear prevention, remediation and compensation costs rather than externalising them to affected communities or the public; it does not purchase permission to pollute.
- **Environmental justice tests distribution and voice:** Environmental justice asks who receives environmental benefits, who bears pollution, displacement and risk, and who has meaningful voice and remedy, with special attention to poor, tribal, nomadic and marginalised communities.
- **Anthropocentric, biocentric and ecocentric reasons differ:** Anthropocentrism values nature chiefly for human interests, biocentrism recognises inherent worth in living beings, and ecocentrism values ecosystems and ecological processes as wholes; administrative reasoning should identify rather than conflate them.
- **CBDR-RC and EIA require differentiated, least-damaging choices:** CBDR-RC joins common climate responsibility with historical contribution and capability, while sensitive border projects require security-aware EIA, community consideration, classified handling where necessary, and the least ecologically damaging feasible alternative.

SOURCE AND ATTRIBUTION DISCIPLINE

The corrected canonical Topic 13 Basic and Advanced owners control the doctrinal sequence. Official question wording and material case facts come from the stated local GS-IV PDFs or their document-order knowledge exports; routing ledgers establish ownership but do not authorise invented facts. Topic 13 owns only the isolated 2024 Q1(a), 2024 Q2(b), 2025 Q1(a) and 2025 Q2(b) subparts: their paired subparts route respectively to Topic 1, Topic 12, Topic 14 and Topic 12. The full 2024 Q7, 2024 Q12 and 2025 Q8 cases retain Topic 22 method cross-links; Topic 12 is only a corporate cross-link for Q7, and research/drug ethics is the specialist cross-link for Q12. Do not add algorithmic discrimination to 2024 Q7. Do not add degraded forest, city-periphery or court-monitoring facts to 2025 Q8. AI is framed as decision support subject to data, design, deployment and feedback bias controls, explainability, named accountability, meaningful human oversight, contestability, hallucination checks, red-teaming, procurement controls and continuing monitoring. Puttaswamy is the constitutional privacy foundation: use legality, legitimate aim, rational connection, necessity or least-restrictive means, proportionate balancing and procedural safeguards rather than citing a useful aim alone. DPDP commencement is phased under G.S.R. 843(E): specified provisions commenced on 13 November 2025, the Consent Manager tranche begins 13 November 2026, and core notice, consent, rights, fiduciary, Board and penalty provisions begin 13 May 2027. Do not call the Act fully operational at the 29 August 2026 cut-off; ethical duties of dignity, autonomy, purpose limitation, minimisation and security apply independently of deferred statutory operation. Rio Principle 15 is the classic environmental precautionary approach. Vellore is an Indian environmental-law authority for sustainable development, precaution and polluter-pays in its tannery-pollution setting; neither Rio nor Vellore is binding AI law. Precaution may guide high-stakes AI only by explicit analogy. Keep precaution ex ante and polluter-pays ex post analytically distinct. Distinguish anthropocentric, biocentric and ecocentric reasons; use environmental justice for burden, voice and remedy; and state CBDR-RC as differentiated responsibility, not zero responsibility. Border-clearance answers require a specific security purpose, EIA and cumulative-risk discipline, community consideration, classified expert review where genuinely necessary, and the least ecologically damaging feasible alternative. The India AI Governance Guidelines and India AI Impact Summit are current official policy anchors, not a comprehensive AI statute or evidence that a particular system is fair, safe or low-carbon.

ANSWER SPINE

1. Define the precise ethical concept or teaching.
2. State a qualified thesis rather than a moral slogan.
3. Explain the causal or decision-making mechanism.
4. Add one specific Indian administrative illustration.
5. Test the strongest limitation or attribution caveat.
6. End with a practical, institution-aware verdict.

EMERGING-ETHICS CORE MAP

- **Shared structure:** AI and environmental harms may be opaque, delayed, diffuse, scalable and borne by people without effective voice.
- **Ethical sequence:** rights and dignity floor -> long-horizon consequences -> virtue and restraint -> precaution before scale -> accountability and remedy.
- **AI verdict:** useful decision support, never an unanswerable moral or legal agent.
- **Environmental verdict:** legitimate development within ecological, distributive and intergenerational limits.

AI DECISION GOVERNANCE

- **Benefit claim:** speed, consistency, pattern detection, auditability and better targeting of scarce administrative attention.
- **Bias chain:** data representation and labels -> proxy and objective design -> deployment context -> feedback loops -> model drift.
- **Explainability:** give affected persons intelligible decisive factors, uncertainty and limits; source-code disclosure alone is not enough.
- **Accountability:** name developer, procurer, deploying department and final decision-maker; preserve logs, audit access, incident duties and sanctions.
- **Human oversight:** reviewer needs competence, time, authority, information and freedom to depart; rubber-stamping is not human-in-the-loop.
- **Contestability:** notice of material automation -> data correction -> simple challenge -> timely human reconsideration -> correction or remedy.
- **Reliability:** verify generated facts and citations; test domain performance; disclose uncertainty; never use unverified hallucinated output alone for consequential action.
- **Human capability:** counter automation bias and de-skilling through training, manual samples, recorded departures and periodic unaided judgment.
- **Procurement:** define purpose and risk -> representative tests -> security and bias red-teaming -> documentation -> audit rights -> change control -> monitoring -> exit.

DIGITAL PUBLIC SPHERE AND INCLUSION

- **Deepfakes:** authenticity and provenance problem; preserve originals, verify, correct rapidly and avoid disproportionate suppression.
- **Social media:** expression and association versus misinformation, manipulation, addictive amplification, privacy invasion, cyberbullying and polarisation.
- **Moderation:** transparent reasons, proportionate rules, appeal and child-sensitive safeguards; neither total laissez-faire nor blanket censorship.
- **Digital divide:** connectivity + affordability + device + language + disability access + literacy + assistance + effective offline alternative.
- **2020 Q5(b):** assess tradition and network culture through constitutional dignity and equality, not romanticise either side.
- **2022 Q4(b):** online efficiency is unethical when essential services become inaccessible to vulnerable persons.
- **2023 Q12:** truthful counter-speech must protect minors, minimise disclosure and prefer institutional remedy over viral escalation.
- **2025 Q1(a):** organise dilemmas as expression/safety, relevance/manipulation, anonymity/accountability, virality/truth and moderation/censorship.

PRIVACY, PUTTASWAMY AND DPDP

- **Privacy:** dignity, decisional autonomy and contextual control over collection, combination, inference, reuse and disclosure.
- **Puttaswamy:** nine-judge bench, 24 August 2017, privacy fundamental under Article 21 and Part III freedoms.
- **Full State restriction test:** legality -> legitimate aim -> rational connection -> necessity or least-restrictive means -> proportionate balance -> procedural safeguards.
- **Data ethics:** meaningful consent + purpose limitation + minimisation + accuracy + retention limits + security + correction + remedy.
- **DPDP chronology:** 13 Nov 2025 specified immediate tranche; 13 Nov 2026 Consent Manager tranche; 13 May 2027 core notice, consent, rights, fiduciary, Board and penalty provisions.
- **Exam limit:** enacted or notified does not mean every duty is presently in force; ethical responsibility precedes full statutory commencement.

AI ENVIRONMENTAL FOOTPRINT

- **Lifecycle:** training electricity + repeated inference + cooling and power water + embodied hardware + mineral extraction + replacement + e-waste.
- **Measure honestly:** report absolute as well as efficiency impacts; renewable electricity alone does not erase water, hardware or waste burdens.
- **Transition controls:** carbon and water budgets, efficient models, utilisation, workload timing, renewable additionality, water-aware siting, repair, reuse and verified e-waste handling.
- **2024 Q7 facts:** ABC is second largest worldwide in the Third World; 2023 GHG emissions were 48% above 2019; AI data-centre energy demand grew; 2030 net-zero goal and competitive pressure remain. No algorithmic-discrimination strand.

ENVIRONMENTAL ETHICS DOCTRINE

- **Anthropocentric:** nature valued chiefly for human interests; useful but may under-protect non-instrumental ecological value.
- **Biocentric:** living beings have inherent worth; difficult conflicts arise where basic human survival is involved.
- **Ecocentric:** value attaches to ecosystem wholes and processes; must not erase sustainable traditional community use.
- **Sustainable development:** present needs without compromising future generations' ability to meet their needs, Brundtland 1987.
- **Intergenerational equity:** present officials represent future persons who cannot vote or contest today's depletion.

- **Precaution:** serious or irreversible risk plus uncertainty justifies proportionate preventive action; Rio Principle 15 uses 'precautionary approach'.
- **Polluter-pays:** polluter bears prevention, remediation and compensation; payment is not a licence to pollute.
- **Environmental justice:** test distribution of benefits and burdens, meaningful voice, recognition, compensation and remedy.
- **Vellore, 1996:** tannery effluent in the Palar setting; sustainable development, precaution and polluter-pays recognised in Indian environmental law. It is not a general environmental code or AI law.
- **CBDR-RC:** common climate duty differentiated by historical contribution and respective capability; developing-country duty is differentiated, not absent.

CLEARANCE, SECURITY AND COMMUNITY

- **Border method:** specify legitimate security objective -> compare routes and technologies -> rigorous EIA and cumulative/disaster risk -> least-damaging feasible alternative.
- **Secrecy discipline:** consult communities and publish non-sensitive reasons; genuinely classified facts may be reviewed by cleared independent experts.
- **Conditions:** seasonal controls, wildlife passages, restoration, monitoring, stop triggers and compliance reporting.
- **2025 Q2(b):** reject both automatic national-security exemption and automatic ecological prohibition.
- **2025 Q8 facts:** age-old trees, medicinal plants, biodiversity, micro-climate and rainfall, wildlife habitat, soil fertility, erosion control, tribal and nomadic livelihoods, housing duty, wildlife conflict and alleged hideouts. Do not invent degradation, city-periphery or court monitoring.

CURRENT OFFICIAL ANCHOR

- **India AI Governance Guidelines:** policy guidance organised through principles, recommendations, action plan and practical guidance, with voluntary commitments and focus on fairness, accountability, safety, inclusivity, transparency, risk classification, oversight, testing and grievance.
- **India AI Impact Summit 2026:** 19-20 February at Bharat Mandapam within the 16-20 February programme; people-centric, inclusive and protect-planet framing; official challenges include employment disruption, bias and energy consumption.
- **Limit:** neither anchor is a comprehensive AI statute or proof of a deployed system's fairness, safety or carbon performance.

TWELVE-PYQ ANSWER SPINES

- **2018 Q10:** relief and evidence -> conditional closure/restart -> polluter-funded restoration -> worker transition -> participatory monitoring.
- **2020 Q5(b):** liberating and disruptive internet values -> rights-based critical adaptation.
- **2021 Q2(a):** digital evidence benefits -> data and context limits -> supervised, appealable use.
- **2022 Q4(b):** online access benefits -> capability divide and privacy -> assisted and offline channels.
- **2023 Q12:** child welfare and evidence -> pros/cons of counter-video -> anonymised correction and institutional remedy.
- **2024 Q1(a):** benefit -> bias, opacity, reliability and de-skilling -> audit, oversight and contestability.
- **2024 Q2(b):** greed and temperance -> intergenerational justice -> sustainable and differentiated transition.
- **2024 Q7:** preserve emissions, energy, net-zero and competition facts -> lifecycle plan and audited milestones.
- **2024 Q12:** refuse manipulation, consent bypass and patent misuse -> escalate -> lawful accelerated research.
- **2025 Q1(a):** map paired social-media dilemmas -> platform, user and state safeguards.
- **2025 Q2(b):** security purpose -> least-damage EIA -> classified review and public reasons.
- **2025 Q8:** dignity and shelter versus complete ecosystem services -> alternatives first -> minimum residual diversion.

FINAL ANSWER METHOD

1. Identify whether the demand is AI, data, social-media, climate, pollution, clearance or combined ethics.

2. State a qualified thesis: innovation or development is legitimate only within rights, ecological limits and accountable institutions.
3. Name the exact mechanism: bias stage, opacity, automation bias, consent failure, precaution, polluter-pays, environmental justice or CBDR-RC.
4. Add one verified Indian constitutional, judicial, statutory, PYQ or current-policy anchor with its date and limit.
5. For cases, map stakeholders including vulnerable communities, non-human ecology and future generations; list options and consequences.
6. Recommend operational safeguards: owner, test, reason, appeal, EIA, alternative, monitoring, remedy and stop trigger.
7. Test the strongest counterpoint: efficiency, secrecy, innovation, livelihood, security or measurement uncertainty.
8. Conclude with a reviewable balance, never a slogan, false statutory claim or invented case fact.